Complex Analysis

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1 Complex Differentiation

1.1 Definitions

The goal of this course is to develop the surprisingly rich theory of complex valued functions of one complex variable and the theory of integrating such functions along complex paths. The motivations for investigating such topics are complex polynomials of interest in geometry and number theory.

We will also look at functions defined by power series such as the map from $s \to \sum_{n \in N} \frac{1}{n^s} = \zeta(s)$ will define a complex differentiable function for $\mathrm{Re}(s) > 1$. There is also a connection to harmonic functions which is developed further in Analysis of Functions. We can also use complex methods to solve classical integrals or and differential equations which is developed more in IB Complex Methods.

For this course we will use $z \in \mathbb{C}$ and $x = \mathrm{Re}(z), y = \mathrm{Im}(z)$. We will also use θ for the argument of z, $\arg(z)$ which is well-defined up to adding $2\pi\mathbb{Z}$. We will use the principal argument $\mathrm{Arg}(z) \in (-\pi, \pi]$.

Definition. (Open disc) An *open disc* or *open ball* centred at a with radius r in $\mathbb{C}$ is the set $\{z \in \mathbb{C} \mid |z - a| < r\} = B(a, r) = D(a, r)$.

Remark. We will use $\mathbb{D} = B(0, 1)$ and $\bar{B}(a, r) = \{z \in \mathbb{C} \mid |z - a| \leq r\}$.

We use $\mathbb{C}^*$ to denote $\mathbb{C} \setminus \{0\}$.

Recall that a set $U \subseteq \mathbb{C}$ is open if it contains an open disc about each of its points.

Definition. (Path) A *path* in $U \subseteq \mathbb{C}$ is a continuous map $\gamma : [a, b] \to U$.

Definition. (Path-connected) We say that $U \subseteq \mathbb{C}$ is *path-connected* if for all $x, y \in U$ there exists a path $\gamma : [0, 1] \to U$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Definition. (Domain) A *domain* in $\mathbb{C}$ is a non-empty path-connected subset of $\mathbb{C}$.

Definition. (Closed path) If γ is a path and $\gamma(a) = \gamma(b)$ then we say that γ is a *closed path*.

Definition. (C^1 path) We say a path is C^1 if it is continuously differentiable. We say a path is *piecewise* C^1 if it has finitely many non-differentiable points but still globally continuous.

Definition. (Simple path) A path is *simple* if it is injective except perhaps at the end-points.

Definition. Let $U \subseteq \mathbb{C}$ be open.

(i) We say that $f : U \to \mathbb{C}$ is *differentiable* at $w \in U$ if

$$f'(w) = \lim_{z \to w} \frac{f(z) - f(w)}{z - w}$$

exists.

(ii) We say that f is *holomorphic* at $w \in U$ if $\exists \varepsilon > 0$ such that f is differentiable on $B(w, \varepsilon) \subseteq U$.

(iii) If f is holomorphic everywhere, we say f is *entire*.

Remark. Some authors use analytic for holomorphic.

Remark. The usual rules for differenting sums and products and the inverse of a function (when it exists) apply exactly like we say in IA Analysis I with exactly the same proof.

Any $f : U \to \mathbb{C}$ can be written as $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ where $u, v : U \to \mathbb{R}$ are the real and imaginary parts of f.

Recall that $u : U \to \mathbb{R}$ is differentiable at $(c, d) \in U$ with derivative $Du \mid_{(c,d)} = (\lambda, \mu)$ if and only if

$$\frac{u(x, y) - u(c, d) - (\lambda(x - c) + \mu(y - d))}{\sqrt{(x - c)^2 + (y - d)^2}} \to 0$$

as $(x, y) \to (c, d)$.

Proposition. (Cauchy-Riemann equations) Let $f : U \to \mathbb{C}$ be defined on an open set U and write $f = u + iv$, then f is differentiable t $w = c + id \in U$ with $f'(w) = p + iq$ if and only if u and w are both differentiable at (c, d) and $u_x = v_y = p$ and $-u_y = v_x = q$ at (c, d). Then $f'(w) = u_x(c, d) + iv_x(c, d)$.

Proof. f is differentiable at w with derivative $p + iq$ if and only if

$$\lim_{z \to w} \frac{f(z) - f(w) - (z - w)(p + iq)}{z - w} = 0$$

One can check that $\lim_{z \to a} \frac{f(z)}{g)z)} = 0$ if and only if $\lim_{z \to a} \frac{f(z)}{|g(z)|} = 0$ and $(p + iq)(z - w) = p(x - c) - q(y - d) + i(q(x - c) + p(y - d))$ so using these and taking real and imaginary parts we get that

$$\lim_{(x,y) \to (c,d)} \frac{u(x, y) - u(c, d) - (p(x - c) - q(y - d))}{\sqrt{(x - c)^2 + (y - c)^2}} = 0$$

for the real part. And

$$\lim_{(x,y) \to (c,d)} \frac{v(x, y) - v(c, d) - (q(x - c) + p(y - d))}{\sqrt{(x - c)^2 + (y - c)^2}} = 0$$

for the imaginary part. This is equivalent to saying that u is differentiable with $Du \mid_{(c,d)} = (p, -q)$ and v is differentiable with $Dv \mid_{(c,d)} = (q, p)$. $\square$

Remark. Let's make some remarks about the Cauchy-Riemann equations.

(i) If $f = u + iv$ and $u_x = v_u$ and $u_y = -v_x$ at a point w we *cannot* conclude that f is differentiable at w (Example Sheet 1).

(ii) If the partial derivatives u_x, u_y, v_x, v_y exist and are continuous in an open neighbourhood of w then the Cauchy-Riemann equations holding does imply complex differentiability.

Let's see some examples.

(i) Polynomials are sums and products of the identity function, hence they are entire.

(ii) If P and Q are polynomials, and $U \subseteq \mathbb{C} \setminus \{x \mid Q(x) = 0\}$ then $\frac{P}{Q}$ is differentiable on U. These are called *rational functions*.

(iii) If $f(x) = |x|$, this is not differentiable anywhere in $\mathbb{C}$. $f = u + iv$ with $u = \sqrt{x^2 + y^2}$ and $v = 0$. If $(x, y) \neq (0, 0)$ then

$$u_x = \frac{x}{\sqrt{x^2 + y^2}}, \qquad u_y = \frac{y}{\sqrt{x^2 + y^2}}.$$

So the Cauchy-Riemann equations do not hold, and if $(x, y) = (0, 0)$ we know that this isn't even differentiable in the real case, hence it's also not differentiable in the complex case. So f isn't differentiable anywhere.

Remark. Later we'll see that if f is holomorphic on U, then f' is also holomorphic on U. Then from the Cauchy-Riemann equations, we can see that f is harmonic. Conversely we can later see that every harmonic function on an open set in $\mathbb{R}^2$ is locally $\mathrm{Re}(f)$ for some holomorphic function f.

1.2 Conformal maps

Proposition. Let $U \subseteq \mathbb{C}$ be a domain and suppose that $f : U \to \mathbb{C}$ is holomorphic and $f'(z) = 0$ on U. Then f is constant.

Proof. We will without proof the following elementary topological fact.

Lemma. If U is a domain and $\gamma : [0, 1] \to U$ is a path with $\gamma(0) = a$ and $\gamma(1) = b$. Then there is another path $\bar{\gamma} : [0, 1] \to U$ with $\bar{\gamma}(0) = a$ and $\bar{\gamma}(1) = b$ where $\bar{\gamma}$ is composed of finitely many segments, each parallel to the x or y axis, so $\bar{\gamma}$ is piecewise-C^1.

Given this we know that $u_x = v_y$ and $u_y = -v_x$ hold. Since f' is zero, all these partials vanish on U. Now the usual mean value theorem shows that u, v are constant along the segments of $\bar{\gamma}$. So $f(a) = f(b)$. $\square$

Definition. (Conformal) If f is holomorphc at a point w and $f'(w) \neq 0$ we say that f is *conformal* at w.

This is a geometric property of f.

Suppose that U is a domain and $f : U \to \mathbb{C}$ is conformal at w. Let's take $\gamma_i : (-\varepsilon, \varepsilon) \to U$ such that $\gamma_i(0) = w$, and $\gamma_i'(0) \neq 0$. So γ_i have non-zero tangent vectors through w. The angle

between these vectors is $\arg(\gamma_1'(0) - \arg(\gamma_2'(0))$. But then

$$\frac{(f\gamma_1)'(0)}{(f\gamma_2)'(0)} = \frac{f'(w)}{f'(w)} = \frac{\gamma_1'(0)}{\gamma_2'(0)} = 1$$

and so the angle between $f\gamma_1$ and $f\gamma_2$ at $f(w)$ the same as for γ_1 and γ_2. So conformal means that f *preserves* angles.

Definition. (Conformal equivalence) If U, V are open in $\mathbb{C}$ and $f : U \to V$ is a holomorphic bijection which is everywhere conformal on U we say that f is a *conformal equivalence* and U and V are conformally equivalent.

Remark. In this setting, if f is conformal, the inverse function theorem says that f is locally invertible and that local inverse is complex differentiable. If f is a bijection is it globally invertible, and hence the chain rule shows that f^{-1} is conformal.

Let's see some examples.

(i) A linear map $f(z) = az + b$ with $a \neq 0$ is a conformal equivalence $\mathbb{C} \to \mathbb{C}$.

(ii) $f(z) = z^n$ takes $\{z \mid 0 \leq \operatorname{Arg}(z) \leq \frac{\pi}{n}\} \to \{z \in \mathbb{C} \mid \operatorname{Im}(z) > 0\} = \mathcal{H}$. This fails at the origin, but if we consider the open sector and open upper half plane.

(iii) The exponential map, $z \to \exp z = e^x e^{iy}$ sends verticle lines to circles with radius $e^{\operatorname{Re} z}$.

(iv) $z \in \mathcal{H} \iff z$ is closer to i than to $-i \iff \left|\frac{z-i}{z+1}\right| < 1$. The Mobius map $z \to \frac{z-i}{z+i}$ takes $\mathcal{H} \to \mathbb{D}$. Moreover $f'(z) = \frac{2i}{(z+i)^2}$ is non-zero for $z \in \mathcal{H}$, so f is an conformal equivalence.

Recall that the Mobius group, $\mathcal{M}$, is the group of mappings $z \to \frac{az+b}{cz+d}$ with $ad - bc \neq 0$. $A \in \mathcal{M}$ defines a conformal equivalence from $\mathbb{C} \setminus \{-\frac{d}{c}\} \to \mathbb{C} \setminus \{\frac{a}{c}\}$, but it's much better to think of it as a conformal equivalence to the extended complex plane $\mathbb{C}_\infty$. Recall that Mobius maps are triply transitive, so

$$z \to \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

sends the triple (z_1, z_2, z_3) to $(0, 1, \infty)$.

Recall further that Mobius maps send circlines to circlines (where a circline is a circle or a line plus the point at infinity).

Suppose we have a sector bounded between lines L_1, L_2. What are the images of a sector under a Mobius map? This depends on if our Mobius map T is such that $T^{-1}(\infty) \in L_1$ or $T^{-1}(\infty) \in L_2$. Any region formed by two non-pallel lines, a circle and a line, or two circles could be image under a Mobius map.

The *Joukowsky transform* is the map sending $z \to \frac{1}{2}\left(z + \frac{1}{z}\right) = \frac{z^2+1}{2z}$. We can compute that $f'(z) = \frac{1}{2} - \frac{1}{2z^2}$ and f is holomorphic except at 0 and is conformal except at 0 and ± 1. Take the circle passing through -1 and $-i$ not centred at the origin. Under the map f, creates a kink at -1 since it's not conformal, but the curve is nice everywhere else since it's conformal on the rest of the circle. The transformed curve resembles an aerofoil. The incompressible fluid equations, $\nabla \times v = 0, v = \nabla\phi$ implies that $\nabla^2\phi$ vanishes, so ϕ is harmonic. So ϕ is locally $\operatorname{Re}(f)$, with f holomorphic. This is used to transfer questions about fluid flow about complicated shapes to the much simplier circle.

Let $S^1 = \partial\mathbb{D} = \{x \in \mathbb{C} : |x| = 1\}$

Definition. (Simply connceted) Let $U \in \mathbb{C}$ be a domain. We say that U is *simply connected* if every continuous map $\gamma : S^1 \to U$ extends to a map $\hat{\gamma} : \overline{\mathbb{D}} \to U$ such that $\hat{\gamma}\,|_{\partial\mathbb{D}} = \gamma$.

Remark. This is the notion that any loop can be continuously shrunk to a point.

For example for the annulus, if we take γ to be the closed path going all the way around the annulus, we will later prove that no such $\hat{\gamma}$ can exist.

Theorem. (Riemann mapping theorem) If $U \subset \mathbb{C}$ is a proper subdomain and U is simply connected, then U is conformally equivalent to the disc, $\mathbb{D} = \{z : |z| < 1\}$.

Remark. The notation of being simply connected is invariant under conformal equivalences. Being simply connected is a topological property and is invariant under any homeomorphism (since conformal maps are homeomorphism).

Remark. The condition that $U \neq \mathbb{C}$ is essential. If $\mathbb{C}$ was conformally equivalent to the open disc then there would exist a function $f : \mathbb{C} \to \mathbb{D}$ conformal, hence a homeomorphism, but from IB Topological Spaces, homeomorphisms are closed maps, hence since $\mathbb{C}$ is closed, $\mathbb{D}$ should be closed, but it isn't hence f cannot exist. We will see a non-topological proof later in the course.

A useful construction of holomorphic functions (and we'll see later the universal construction) is convergent power series.

1.3 Power series

Let's see some reminders from IB Analysis II.

(i) A sequence f_n of functions on a set T converges uniformly to a function f on T if that $\forall \varepsilon > 0\ \exists N$ s.t. $\forall\ n \geq N \in \mathbb{N},\ |f_n(x) - f(x)| < \varepsilon$.

(ii) The uniform limit of continuous functions is continuous.

(iii) The Weiestress M-test says that if we have values $M_n \in \mathbb{R}_{>0}$ such that $|f_n(x)| \leq M_n$ for all $x \in T$, then if $\sum_{n=1}^{\infty} M_n$ converges, then the series $\sum_{n=1}^{\infty} f_n(x)$ converges absolutely and uniformly on T.

(iv) Suppose we have a sequence $\{a_n\}$ then there is a unique $R \in [0, \infty]$ such that the series $\sum_{n=0}^{\infty} a_n(z-a)^n$ converges absolutely if $|z-a| < R$ and diverges if $|z-a| > R$. Moreover if $0 < r < R$ then the series converges uniformly on $\{z : |z-a| < r\}$. This is called the radius of convergence. The root test says that

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}}.$$

Theorem. Suppose that

$$f(z) = \sum_{n=0}^{\infty} c_n(z-a)^n$$

is a complex power series with radius of convergence $R > 0$. Then

(i) f is holomorphic on $B(a, R) = \{z : |z - a| < R\}$;

(ii) $f'(z) = \sum nc_n(z - a)^{n-1}$ which also has radius of convergence R;

(iii) f is infinitely complex differentiable on $B(a, R)$ with

$$f^{(n)}(a) = n!c_n.$$

Proof. We can see (iii) follows from (ii) so enoughto prove (i) and (ii). Take $a = 0$ *wlog*. Since $n|c_n| > |c_n|$ the radius of convergence of $\sum_{n>=1} nc_n(x_a)^{n-1}$ is at most R. But if $|z| < R_1 < R$ then we can see that

$$\frac{|nc_n z^{n-1}|}{|c_n R_1^{n-1}|} = n \left| \frac{z}{R_1} \right|^{n-1} \to 0$$

which is true for all $R_1 < R$, hence the radius of convergence must be R.

Now to show f is differentiable with that derivative. Pick z, w with $|z|, |w| \leq R_1 < R$. Define

$$\varphi(z, w) = \sum_{n=1}^{\infty} c_n \sum_{j=0}^{n-1} z^j w^{n-1-j}.$$

We can see that

$$\left| c_n \sum_{j=0}^{n-1} z^j w^{n-1-j} \right| \leq n|c_n|R_1^n$$

so this convergences uniformly with $|z| \leq R_1$ and $|w| < R_1$ so the limit is continuous. If $z \neq w$ we can sum as a geometric series, so

$$\varphi(z, w) = \sum_{n=1}^{\infty} c_n \left(\frac{z^n - w^n}{z - w} \right) = \frac{f(z) - f(w)}{z - w}.$$

Taking the limit as $w \to z$ we get that

$$\begin{aligned} \lim_{w \to z} \frac{f(z) - f(w)}{z - w} &= \varphi(z, z) \\ &= \sum_{n=1}^{\infty} c_n n z^{n-1} = f'(z). \end{aligned}$$

Hence we're done. □

Corollary. Suppose we have a power series

$$f(z) = \sum_{n=-0}^{\infty} c_n (z - a)^n$$

with $R > 0$ and suppose that f vanishes on $B(a, \varepsilon)$ with $\varepsilon \in (0, R)$. Then f vanishes identically.

Proof. All derivatives are zero at $x = a$, hence $f(z) = 0$ □.

Now we can define some familiar functions

1.4 Exponentials, logarithms, and branch cuts

Definition. (Exponential function) We define the function

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

which has the properties

(i) Radius of convergence is ∞;
(ii) $\frac{d}{dz} e^z = e^z$;
(iii) $e^0 = 1$. If we fix w and let $F(z) = e^{z+w} e^{-z}$ then $F'(z)$ vanishes, hence F is constant, and $F(0) = e^w$, so $e^{z+w} = e^z e^w$ holds for all $z, w \in \mathbb{C}$.
(iv) e^z never vanishes on $\mathbb{C}$.

Remark. We define the trigonometic functions again as we did in $\mathbb{R}$. The exponential function, sin and cos are entire.

This function is invertible over $\mathbb{R}$, but in $\mathbb{C}$ it is not invertible. In fact for every $z \in \mathbb{C}$ there exists infinitely many $w \in \mathbb{C}$ distinct such that $e^w = z$. This can be seen since exp is $2\pi i$ peroidic.

Definition. (Branch of the logarithm) Let $U \subseteq \mathbb{C}^* = \mathbb{C} \setminus \{0\}$ open. A continuous function $\lambda : U \to \mathbb{C}$ is *branch of the logarithm* on U if $e^{\lambda(z)} = z$ for all $z \in U$.

The classical example is $U = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, the slit plane. Let $\mathrm{Log} : U \to \mathbb{C}$ be the defined by $\mathrm{Log}(z) = \log(z) + i\theta$ where $\theta = \mathrm{Arg}(z)$. This is called the principal branch of the logarithm.

Proposition. Log is holomorphic on $\mathbb{C} \setminus \mathbb{R}_{\leq 0}$ with

(i) $\frac{d}{dz} \mathrm{Log}(z) = \frac{1}{z}$;
(ii) If $|z| < 1$ then

$$\mathrm{Log}(1+z) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n}.$$

Proof. Log is continuous on U and exp is continuous on $\mathbb{C}$. If $z = e^x$ and $w = e^y$, then

$$\frac{\mathrm{Log}(z) - \mathrm{Log}(w)}{z - w} = \frac{x - y}{e^x - e^y}$$

which as $x \to y$ converges to $\frac{1}{e^y} = \frac{1}{w}$.

By the ratio test, $\mathrm{Log}(1+z)$ has a radius of convergence 1. We know that $\mathrm{Log}(1+z)$ has derivative

$$1 - z + z^2 - \cdot = \frac{1}{1+z}$$

so since the claimed power series and $\mathrm{Log}(1+z)$ have the same derivative, they differ by a constant, and since they are both zero at $z = 0$, that constant is zero, so $\mathrm{Log}(1+z)$ has the power series as claimed. $\square$

Remark.

(i) Later we'll see that there is no branch of the logarithm defined on $\mathbb{C}^*$.

(ii) You can define a continuous branch of log on any simply connected domain not containing zero.

Definition. (Multivalued power function) For a value $\alpha \in \mathbb{C}$, the multivalued function z^α is by definition $\exp(\alpha \log(z))$..

Remark. If for $U \subseteq \mathbb{C}^*$ we have a branch of log specified on U, then z^α becomes single-valued on U.

Remark. If $\alpha \in \mathbb{Z}$, then this is single-valued and if $\alpha \in \mathbb{Q}$ then it is finitely many valued, for example $z^{1/2}$ has two values. Note we cannot define a square root function globally on $\mathbb{C}$.

The function $f(z) = z(z-1)$ admits a single valued square root on each of the domains $\mathbb{C} \setminus [0,1]$ and $\mathbb{C} \setminus (\mathbb{R}_{\geq 0} \cup \mathbb{R}_{\geq 1})$. Hence

$$(z(z_1))^{\frac{1}{2}} = \exp\left(\frac{1}{2}(\log(z) - \log(z-1))\right).$$

So taking a path around the removed interval in the complex plane is fine since both log terms jump, giving a total jump $2\pi i$ which doesn't change the value since exp is $2\pi i$ periodic.

Definition. (Branch point) A point $p \in \mathbb{C}$ is a *branch point* of a multivalued function ϕ if there is no continuous single-value definition of ϕ in $B(0, \varepsilon)$ for any $\varepsilon > 0$.

For example 0 is a branch of log. $\{0, 1\}$ are branch points of $(z(z-1))^{1/2}$.

2 Contour Integration

2.1 Basic properties

Definition. (Riemann integrablility in $\mathbb{C}$) A function $f : [a, b] \to \mathbb{C}$ is *Riemann integrable* if it's real and imaginary parts are both Riemann integrable and

$$\int_a^b f(t)\mathrm{d}t = \int_a^b \mathrm{Re}(f(t))\mathrm{d}t + i \int_a^b \mathrm{Im}(f(t))\mathrm{d}t.$$

Note that we have

$$\left| \int_a^b f(t)\mathrm{d}t \right| \leq \int_a^b |f(t)|\mathrm{d}t.$$

From now on f will always be continuous and hence Riemann integrable.

Proposition. For $f : [a, b] \to \mathbb{C}$ we have that

$$\left| \int_a^b f(t)\mathrm{d}t \right| \leq \sup_{t \in [a,b]} |f(t)|(b-a)$$

with equality if and only if f is constant.

Proof. Let $\theta = \operatorname{Arg} \int_a^b f(t)\mathrm{d}t$ and $M = \sup_t |f(t)|$. Hence

$$\begin{aligned}\left|\int_a^b f(t)\mathrm{d}t\right| &= e^{-i\theta}\int_a^b f(t)\mathrm{d}t \\ &= \int_a^b \operatorname{Re}(e^{-i\theta}f(t))\mathrm{d}t \\ &\leq \int_a^b |e^{-i\theta}f(t)|\mathrm{d}t \\ &\leq M(b-a)\end{aligned}$$

and equality is equivalent to $\operatorname{Arg}(\theta)$ being constant and $|f| = M$ everywhere, hence equality occurs if and only if f is constant. □

Definition. (Contour) A *contour* is a simple, closed path.

Definition. (Contour integral) If U is a domain, $f : U \to \mathbb{C}$ is continuous and $\gamma : [a, b] \to U$ is a C^1-smooth curve then

$$\int_\gamma f(z)\mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\mathrm{d}t.$$

This definition extends in the obvious way to paths. We have some properties about contour integrals.

(i) They are linear;

(ii) They are additive along paths;

(iii) They are independent of parameterisation.

Remark. If we set the length of γ as

$$|\gamma| = \int_a^b |\gamma'(t)|\mathrm{d}t$$

this is *not* independent of parameterisation.

If $(-\gamma)(t) = \gamma(-t)$ so $-\gamma : [-b, -a] \to U$, $\int_{-\gamma} f = -\int_\gamma f$.

Let $U = C^*$ and $f(z) = z^n$, $n \in \mathbb{Z}$. Let $\phi : [0, 2\pi] \to U$ sending $t \to e^{it}$ then

$$\int_\phi f(z)\mathrm{d}z = \begin{cases} 2\pi i & n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Let's prove this.

$$\begin{aligned}\int_\phi f(z)\mathrm{d}z &= \int_0^{2\pi} e^{nit}(ie^{it})\mathrm{d}t \\ &= i\int_0^{2\pi} e^{i(n+1)t}\mathrm{d}t \\ &= \begin{cases} 2\pi i & n+1=0 \\ 0 & \text{otherwise} \end{cases}.\end{aligned}$$

Theorem. (Fundamental Theorem of Calculus) Suppose that U is a domain $F : U \to \mathbb{C}$ is holomorphic and $F'(z)$ is continuous on U. Then for a path $\gamma : [a, b] \to U$

$$\int_\gamma F'(z)\mathrm{d}z = F(\gamma(b)) - F(\gamma(a)).$$

In particular if γ is closed then we get zero.

Proof.

$$\begin{aligned}\int_\gamma F'(z)\mathrm{d}z &= \int_a^b F'(\gamma(t))\gamma'(t)\mathrm{d}t \\ &= \int_a^b (F\circ\gamma)'(t)\mathrm{d}t \\ &= (F\circ\gamma)(b) - (F\circ\gamma)(a) \quad \square\end{aligned}$$

If $n \neq -1$ then $z^n = \frac{d}{dt}\left(\frac{z^{n+1}}{n+1}\right)$ so the integral of z^n around a closed path around the origin, is zero.

Definition. (Antiderivative) If U is a domain and $f : U \to \mathbb{C}$ is continuous and $F : U \to \mathbb{C}$ is holomorphic with $F'(z) = f(z)$ then we say that F is an *antiderivative* for f on U.

So $f(z) = \frac{1}{z}$ has no antiderivative on $\mathbb{C}^*$.

Remark. Later we'll show (without being circular) that the hypothesis saying $F'(z)$ is continuous actually always holds (which we already know if our holomorphic function came from a power series).

2.2 Cauchy's theorem

Lemma. If γ is a C^1 path and f is continuous then

$$\left|\int_\gamma f(z)\mathrm{d}z\right| \leq \mathrm{length}(\gamma)\sup_\gamma |f|.$$

Proof.

$$\begin{aligned}\left|\int_\gamma f(z)\mathrm{d}z\right| &= \left|\int_a^b f(\gamma(t))\gamma'(t)\mathrm{d}t\right| \\ &\leq \int_a^b |f(\gamma(t))||\gamma'(t)|\mathrm{d}t \\ &\leq \sup_{a\in[a,b]} |f(\gamma(t))|\mathrm{length}(\gamma). \quad \square\end{aligned}$$

We can think of the following theorem as a converse to FTC.

Theorem. Let U be a domain and $f : U \to \mathbb{C}$ be continuous. Assume that

$$\int_\gamma f(z)\mathrm{d}z = 0$$

for all piecewise-C_1 closed curves γ in U. Then f has an antiderivative $F(z)$ on U.

Proof. Pick $a_0 \in U$. U is a domain so path-connected. So for $w \in U$ pick a piecwise-C^1 path γ_w from a_0 to w. Set

$$F(w) = \int_{\gamma_w} f(z)\mathrm{d}z.$$

Our hypothesis that the integral over a closed path of f is zero implies that $F(w)$ does not depend on the choice of path hence F is well-defined. U is open so for some $\varepsilon > 0$, $B(w, \varepsilon) \subseteq U$. If $|h| < \varepsilon$ let δ_h be the radial path inside this ball from w to $w + h$. Then

$$\begin{aligned}F(w+h) &= \int_{\gamma_w+\delta_h} f(z)\mathrm{d}z \\ &= \int \gamma_w f(z)\mathrm{d}z + \int_{\delta_h} f(z)\mathrm{d}z \\ &= F(w) + hf(w) - \int_{\delta_h} f(z) - f(w)\mathrm{d}z\end{aligned}$$

So

$$\begin{aligned}\left|\frac{F(w+h)-F(w)}{h} - f(w)\right| &= \left|\frac{1}{h}\int_{\delta_h} f(z) - f(w)\mathrm{d}z\right| \\ &\leq \frac{1}{|h|}\mathrm{length}(\delta_h) \sup_{z\in\delta_h} |f(z) - f(w)|\end{aligned}$$

As $h \to 0$ the right hand side goes to zero by continuity of f. So F is complex differentiable at w and $F'(w) = f(w)$ for all $w \in U$. So f has an antideriviative. $\square$

Definition. (Convex) A domain U is *convex* if $\forall p, q \in U$, the straight line segment connecting p and q lies in U. i.e.

$$pt + (1-t)q \in U, \quad \forall t \in [0,1].$$

Definition. (Star-shaped) A domain U is *star-shaped* if there exists a p_0 such that for all $q \in U$, the line segment p_0, q lies in U.

We have the implications

$$\text{disc} \implies \text{convex} \implies \text{star-shaped} \implies \text{simply connected} \implies \text{path-connected}$$

Remark. All the theorem's we will prove for star-shaped sets related to Cauchy's theorem will extend to further sets. Simply connected is the property for Cauchy's theorem to hold for all curves in a domain, but we'll just prove the theorems for star-shaped sets for now.

Lemma. Let U be star-shaped and $f : U \to \mathbb{C}$ be continuous and suppose that

$$\int_\gamma f = 0$$

where γ is a *triangle* in U where a *triangle* is a piecewise-C_1 path formed by 3 edges of a Euclidean triangle and the whole triangle including its interior is contained in U. Then f has an antiderivative on U.

Proof. Suppose that U is star-like with respect to some point p_0 and for $w \in U$ let γ_w be the path connecting p_0 to w in U. Taking h as before, the paths γ_w, γ_h, γ_{w+h} form a triangle in U so we can use our previous proof.

Theorem. (Cauchy's theorem for triangles) Let U be a domain and $T \in U$ be a triangle in U. If $f : U \to \mathbb{C}$ is holomorphic then

$$\int_{\partial T} f = 0$$

where ∂T is the piecewise-C_1 closed path given by the boundary of T.

Proof. Let $L = \text{length}(\partial T)$ and $I = \left|\int_{\partial T} f\right|$. By bisecting edges we can T into 4 smaller triangles creating all 4 Δ anticlockwise, note that

$$\partial T = \partial \triangle_0 \cup \partial \triangle_1 \cup \partial \triangle_2 \cup \partial \triangle_3$$

so there exists an i such that $\left|\int_{\partial \triangle_i} f\right| \geq \frac{I}{4}$. Label by T_1 such a triangle $\triangle_i$. Now iterate constructing a sequence of triangles $T \supseteq T_1 \supseteq T_2 \supseteq \dots$ hence

(i) $\left|\int_{\partial T_n} f\right| \geq \frac{I}{4^n}$;

(ii) $\text{length}(\partial T_n) = 2^{-n} L$.

Claim. We have that:

$$\bigcap_{n \geq 1} T_n = \{w\}.$$

Proof. We have that the length of the triangle goes to zero hence the diameter goes to zero, so by uniqueness of limits there can be at most one point in the intersection. The intersection being non-empty is a consequence of compactness and follows from applying Bolzano-Weiestrass to each coordinate. So the claim is true.

Define

$$g(z) = \frac{f(z) - f(w)}{z - w} - f'(w)$$

which vanishes at w since f is holomorphic.

$$\begin{aligned}
\frac{I}{4^n} \leq \left| \int_{\partial T_n} f \right| &= \left| \int_{\partial T_n} f(z) - f(w) - (z - w) f'(w) \, \mathrm{d}z \right| \\
&\leq \left| \frac{L}{2^n} \sup_{z \in \partial T_n} |(z - w) g(z)| \right| \\
&\leq \frac{L^2}{4^n} \sup_{z \in \partial T_n} |g(z)| \\
I \leq L^2 \sup_{z \in \partial T_n} |g(z)| &\to 0
\end{aligned}$$

as $n \to \infty$. □

Corollary. (Convex Cauchy) Let f be a holomorphic on a star-shaped domain U. Then $\int_\gamma f = 0$ for any piecewise-C_1 closed curve γ.

Proof. The previous argument implies that $\int_{\partial T} f = 0$ where T is a triangle on U. Thus f has an antiderivative F on U. So $F'(z) = f(z)$. So then

$$\int_\gamma f(z) \mathrm{d}z = \int_\gamma F'(z) \mathrm{d}z = 0$$

by FTC. □

Proposition. Let U be a domain and $f : U \to \mathbb{C}$ be continuous. Suppose that there exists a finite set $S \subset U$ such that f is holomorphic on $U \setminus S$. THen if $T \subseteq U$ is a triangle, then

$$\int_{\partial T} f = 0.$$

Proof. Without loss of generality suppose that S is singleton, subdividing T if necessary. If T contains no points of S we can use Cauchy for triangles. If it contains more than one point of S divide into smaller triangles each containing one or no points and adding. Now suppose that $a \in S$ and $a \in T' \subseteq T$, a smaller triangle. Through a geometric argument we can subdivide T into smaller triangles one of which is T' containing a. Orienting all of these triangles anticlockwise we see that

$$\int_{\partial T} f = \pm \int_{\partial T'} f$$

due to Cauchy's theorem, the integral around all of the other triangles is zero. Now

$$\begin{aligned}\left|\int_{\partial T'} f\right| &= \text{length}(\partial T') \sup_{z\in\partial T'} |f(z)| \\ &= \text{length}(\partial T') \sup_{z\in T} |f(z)|.\end{aligned}$$

Now since $\sup_{z\in T} |f(z)|$ is bounded, we can send $\text{length}(\partial T') \to 0$ by choosing T' to be as small as we like. Hence the integral vanishes. □

Corollary. If U is star-shaped and $S \subset U$ is finite and $f : U \to \mathbb{C}$ is continuous and holomorphic on $U \setminus S$ then

$$\int_\gamma f = 0$$

for all closed curves $\gamma \in U$.

Proof. Using the antiderivative theorem, just as we used before to deduce convex Cauchy from Cauchy for triangles. □

2.3 The Cauchy integral formula

Theorem. Let $U \in \mathbb{C}$ be a domain, $f : U \to \mathbb{C}$ be holomorphic on U and $\overline{B(a,r)} \subseteq U$. For all $z \in B(a,r)$ and $0 < \rho \leq r$, $|z-a| < \rho$, we have that

$$f(z) = \frac{1}{2\pi i}\int_{|w-a|=\rho} \frac{f(w)}{w-z}\mathrm{d}w.$$

Proof. Let

$$g(w) = \begin{cases} \frac{f(w)-f(z)}{w-z} & w \neq z \\ f'(z) & w = z \end{cases}.$$

This is continuous on $\overline{B(a,r)}$ and holomorphic except perhaps at z. So by Cauchy,

$$\int_{\partial B(a,r)=|w-a|=\rho} g(w)\ \mathrm{d}w = 0,$$

hence

$$\int_{|w-a|=\rho} \frac{f(w)}{w-z}\mathrm{d}w = \int_{|w-a|=\rho} \frac{f(z)}{w-z}\ \mathrm{d}w.$$

We'll use that

$$\frac{1}{w-z} = \frac{1}{w-a}\frac{1}{1-\left(\frac{z-a}{w-a}\right)} = \frac{1}{w-a}\sum_{n=0}^{\infty}\left(\frac{z-a}{w-a}\right)^n$$

which converges uniformly on $|w-a| = \rho$. So

$$\begin{aligned}\int_{|w-a|=\rho} \frac{f(w)}{w-z}\ \mathrm{d}w &= f(z)\int_{|w-a|=\rho}\sum_{n=0}^{\infty}\frac{(z-a)^n}{(w-a)^{n+1}}\ \mathrm{d}w \\ &= f(z)\sum_{n=0}^{\infty}\int_{|w-a|=\rho}\frac{(z-a)^n}{(w-a)^{n+1}}\ \mathrm{d}w\end{aligned}$$

swapping the summation and the integral using uniform convergence. Recall that we showed that

$$\int_{|z|=1} z^n \, \mathrm{d}z$$

gave us something nontrivial only when $n = -1$. So by a change of variables from $w - a \to w$ we see only the $n = 0$ term contributes and gives $2\pi i$. Hence we have that

$$\int_{|w-a|=\rho} \frac{f(w)}{w-z} \, \mathrm{d}w = 2\pi i f(z)$$

which gives the result. □

Corollary. (Mean value property) If $f : U \to \mathbb{C}$ is holomorphic on a domain U and $\overline{B(a,r)} \subseteq U$, then

$$f(a) = \int_0^1 f(a + re^{2\pi i t}) \, \mathrm{d}t.$$

So the value of f at a is the average of its values over a small linking circle.

Proof. Parameterise the circle in the Cauchy integral formula, so

$$f(a) = \frac{1}{2\pi i} \int_{|w-a|=r} \frac{f(w)}{w-a} \, \mathrm{d}w = \frac{1}{2\pi i} \int_\gamma \frac{f}{(}\gamma(t))\gamma(t) - a\gamma'(t) \, \mathrm{d}t$$

and the result follows simply by substituing γ in.

Corollary. (Local maximum principle) If $f : B(a,r) \to \mathbb{C}$ is holomorphic and $|f(z)| \leq |f(a)|$ for all $z \in B(a,r)$ then f is constant.

Proof. For $0 < \rho < r$,

$$\begin{aligned} |f(a)| &= \left| \int_0^1 f(a + \rho e^{2\pi i t}) \, \mathrm{d}t \right| \\ &\leq \sup_{|z-a|=\rho} |f(z)| \\ &\leq |f(a)| \end{aligned}$$

So the inequalities are equalties, hence $|f(z)| = |f(a)|$ for all z such that $|z-a| = \rho$, and $\rho \in (0,r)$ is arbitrary hence $|f|$ is constant so by Example Sheet 1 since f is holomorphic, f is constant. □

Theorem. (Liouville's Theorem) Let $f : \mathbb{C} \to \mathbb{C}$ be entire. Then if f is bounded it is constant.

Proof. Let $z_1, z_2 \in C$ and pick $R > \max\{2|z_1|, 2|z_2|\}$. So

$$f(z_1) - f(z_2) = \frac{1}{2\pi i} \int_{\partial B(0,R)} \left(\frac{f(w)}{w - z_2} - \frac{f(w)}{w - z_2} \right) \, \mathrm{d}w.$$

Taking the absolute value of both sides,

$$\begin{aligned}|f(z_1) - f(z_2)| &= \frac{1}{2\pi}\left|\int_{\partial B(0,R)} \frac{f(w)(z_1 - z_2)}{(w - z_1)(w - z_2)}\,\mathrm{d}w\right| \\ &\leq \frac{1}{2\pi} 2\pi R M \frac{|z_1 - z_2|}{(R/2)^2} \to 0\end{aligned}$$

as $R \to \infty$. Hence f is constant. □

Corollary. (Fundamental theorem of algebra) Every non-constant complex polynomial has a root in $\mathbb{C}$.

Remark. This is equivalent to saying all non-constant complex polynomials factors as a product of linear factors.

Proof. Let $p(z) = a_0 + a_1 z + \cdots + a_d z^d$, $a_d \neq 0$ and $d > 0$. Note that

$$\left|\frac{p(z)}{z^d}\right| \to |a_d|$$

as $z \to \infty$. So $|p(z)| \to \infty$ as $|z| = \infty$. Let $f(x) = \frac{1}{p(z)}$. If p never vanishes, then f is entire. Note that $|f(z)| \to 0$ as $|z| \to \infty$, so there exists a $R > 0$ such that $|f(z)| \leq 1$ for $|z| > R$. But f is continuous, so f is bounded on $|z| \leq R$. Hence f is bounded on $\mathbb{C}$, and entire, so we must have that f is constant by Liouville's theorem, which implies that $p(z)$ is constant, which is a contradiction, so p must have a root in $\mathbb{C}$. □

2.4 Taylor and Morera's theorem

Theorem. (Taylor's Theorem) Let $f : B(a, r) \to \mathbb{C}$ be holomorphic. Then f has a convergent power series representation

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n$$

with

$$c_n = \frac{f^{(n)}(a)}{n!} = \frac{1}{2\pi i}\int_{\partial B(a,\rho)} \frac{f(w)}{(w - a)^{n+1}}\,\mathrm{d}w$$

for any $0 < \rho < r$.

Proof. For $0 < \rho < r$, and z such that $|z - a| < \rho$, the Cauchy integral formula gives that

$$f(z) = \frac{1}{2\pi i}\int_{\partial B(a,\rho)} \frac{f(w)}{w - z}\mathrm{d}w.$$

We'll expand

$$\frac{1}{w - z} = \frac{1}{(w - a)\left(1 - \frac{(z-a)}{w-a}\right)}$$

as a power series, so we get that

$$\begin{aligned} f(z) &= \frac{1}{2\pi i}\int_{\partial B(a,\rho)} f(w)\sum_{n=0}^{\infty}\frac{(z-a)^n}{(w-a)^{n+1}}\,\mathrm{d}w \\ &= \sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\int_{\partial B(a,\rho)}\frac{f(w)}{(w-a)^{n+1}}\,\mathrm{d}w\right)(z-a)^n. \end{aligned}$$

And Taylor's theorem follows from here. □

Remark. If $f : U \to \mathbb{C}$ is holomorphic on a domain U, at $z \in U$, there exists a $r > 0$ such that $B(a, r) \subseteq U$, and Taylor's theorem gives us a power series representation on $B(a, r)$, but this may not converge everywhere on U.

Corollary. If f is holomorphic on a domain U, then f is infinitely differentiable on U.

Proof. For each $a \in U$, apply Taylor's theorem for a ball around a. Then since all power series are infinitely differentiable, we have that f is infinitely differentiable at a, hence f is infinitely differentiable on U. □

Theorem. (Morera's theorem) Let U be a domain and $f : U \to \mathbb{C}$ be continuous. If

$$\int_\gamma f = 0$$

for all piecewise-C^1 closed paths in U, the f is holomorphic on U.

Proof. By the hypothesis, we know that f has an antiderivative, F for some F holomorphic. And now we know that the derivative of a holomorphic function is holomorphic, so f is holomorphic. □

Corollary. Let $f_n : U \to \mathbb{C}$ be a sequence of holomorphic functions and suppose $f_n \to f$ uniformly. Then f is holomorphic on U and

$$f'(z) = \lim_{n\to\infty} f_n'(z)$$

for $z \in U$.

Proof. We can say *wlog* $U = B(z, r)$ is a disc. If γ is a closed curve in this disc, then

$$\int_\gamma f_n = 0$$

by convex Cauchy and

$$\int_\gamma f_n \to \int_\gamma f$$

by uniform convergence in IB Analysis. So

$$\int_\gamma f = 0$$

hence since f is continuous (it's the uniform limit of continuous functions) by Morera's theorem it is holomorphic. Then for $0 < \rho < r$, we have that

$$f'(z) = \frac{1}{2\pi i} \int_{\partial B(z,\rho)} \frac{f(w)}{(w-z)^2} \, \mathrm{d}w$$

by Taylor's theorem. And for each $f_n'(z)$ we have similar. Hence

$$\begin{aligned} |f'(z) - f_n'(z)| &= \frac{1}{2\pi} \left| \int_{\partial B(z,\rho)} \frac{f(w) - f_n(w)}{(w-z)^2} \, \mathrm{d}w \right| \\ &\leq \frac{1}{2\pi} 2\pi\rho \frac{1}{\rho^2} \sup_{|w-z|=\rho} |f_1(w) - f_n(w)| \end{aligned}$$

and ρ is fixed so as $n \to \infty$ we get that the difference tends to zero by uniform convergence. □

Corollary. If U is a domain, $f : U \to \mathbb{C}$ is continuous and for some finite set $S \subseteq U$, f is holomorphic on $U \setminus S$, then f is holomorphic on U.

Proof. If $a \in S$ then there exists an $\varepsilon > 0$ such that $B(a, \varepsilon) \subseteq U$ and $B(a, \varepsilon) \cap S = \{a\}$. Cauchy on a disc gives that

$$\int_\gamma f = 0$$

for all closed piecewise-C^1 paths inside $B(a, \varepsilon)$ and f is continuous so by Morera's theorem, f is holomorphic on $B(a, \varepsilon)$. Let's recap some properties of holomorphic functions:

(i) We have the fundemental theorem of calculus,

$$\int_\gamma F' = F(\gamma(b)) - F(\gamma(a)).$$

(ii) Cauchy's theorem for triangles,

$$\int_{\partial T} f = 0$$

if $T \subseteq U$ is a triangle.

(iii) Star-shaped Cauchy,

$$\int_\gamma f = 0$$

for all closed γ in a starlike domain.

(iv) Taylor's theorem gives that f is differentiable and has a local power series expansion.

And some criteria for holomorphicity:

(i) We have the converse fundemental theorem of calculus. If

$$\int_\gamma f = 0$$

for all closed γ, then $f = F'$ for F holomorphic.

(ii) If U is star-shaped, then FTC converse holds if

$$\int_{\partial T} f = 0$$

for all $T \subseteq U$.

(iii) Morera's theorem, if

$$\int_{\gamma} f = 0$$

for all closed γ, then f is holomorphic.

2.5 Homotopy

We should note that Cauchy's theorem holds on a much larger set of domains.

Theorem. (Cauchy for simply connected domains) If $U \subseteq \mathbb{C}$ is a simply connected domain and $f : U \to \mathbb{C}$ is holomorphic then

$$\int_{\gamma} = 0$$

for all closed piecewise-C^1 curves γ in U.

First let's see a definition.

Definition. (Homotopic) Let ϕ, ψ be closed paths in a domain U, say $\phi, \psi : [a, b] \to U$. We say that ϕ, ψ are *homotopic* if there exists a $\Phi : [a, b] \times [0, 1] \to U$ continuous such that $\Phi \mid_{[a,b]\times\{0\}} = \phi$ and $\Phi \mid_{[a,b]\times\{1\}} = \psi$ and for all t $\Phi \mid_{[a,b]\times\{t\}}$ is a closed path.

By definition U is simply connected is every loop is homotopic to a constant loop. We will assume the following claim.

Claim. U is simply connected if whenever ϕ is a piecewise-C^1 closed path, there exists a homotopy Φ to the constant path such that for all t, $\Phi \mid_{[a,b]\times\{t\}}$ is a piecewise-C^1 closed path.

We will assume this without proof.

Proposition. If U is a domain, ψ, ϕ are homotopic piecewise-C^1 closed paths and if $f : U \to \mathbb{C}$ is holomorphic, then

$$\int_{\phi} f = \int_{\psi} f.$$

Proof. Let $\phi, \psi : [a, b] \to U$ be close (piecewise-C^1) paths. We say ϕ and ψ are *elementary deformations* of one another if there exists $a = x_0 < x_1 < \cdots < x_n = b$ and convex open sets $C_i \subseteq U$ for $1 \leq i \leq n$ such that $\underbrace{\phi \mid_{[x_{i-1},x_i]}}_{\phi_i}$ and $\underbrace{\psi \mid_{[x_{i-1},x_i]}}_{\psi_i}$ have images inside C_i. Let $\gamma_i \in C_i$ be the straight path connecting $\phi(x_i)$ to $\psi(x_i)$ which is contained in C_i due to convexity.

Let $\Gamma_i = \phi_i + \gamma_i - \psi_i - \gamma_{i-1}$ which is a closed piecewise-C^1 loop inside C_i. Apply convex Cauchy so

$$\int_{\Gamma_i} f = 0,$$

hence

$$\sum_{i=1}^{n} \int_{\Gamma_i} f = 0.$$

The γ_i all appear twice with opposite orientation, so $\int_\phi f + \int_{-\psi} f = 0$ so

$$\int_\phi f = \int_\psi f$$

if ϕ and ψ are elementary deformations of each other.

Now it is enough to prove that if $\phi \cong \psi$ then there exists $\phi = \phi_0, \phi_1, \ldots, \phi_N = \psi$ such that ψ_{i+1} is an elementary deformation of ϕ_i. Let $\Phi : [a,b] \times [0,1] \to U$ be the homotopy function from $\phi = \Phi_0$ to $\psi = \Phi_1$. The domain of Φ is closed and bounded, hence $\Phi([a,b] \times [0,1])$ is closed and bounded. Hence there exists an ε such that $\operatorname{dist}(\Phi([a,b] \times [0,1]), \mathbb{C} \setminus U) = 2\varepsilon > 0$. So for all $(s,t) \in [a,b] \times [0,1]$, $B(\Phi(s,t), \varepsilon) \subset U$. Now Φ is uniformly continuous so $\exists \delta > 0$ such that

$$|(s,t) - (s',t')| < \delta \implies |\Phi(s,t) - \Phi(s',t')| < \varepsilon.$$

Pick n such that $\frac{1+(b-a)}{n} < \delta$ and set $x_j = a + \frac{(b-a)j}{n}$ for $j = 1, \ldots, n$. Set $\phi_o = \Phi\mid_{[a,b]\times i/n} = \Phi_{i/n}$ so $\phi = \phi_0, \phi_1, \ldots, \phi_n = \psi$ and let $C_{ij} = B(\Phi(x_j, i/n), \varepsilon)$ which is convex and contained in U. Now if $s \in [x_{j-1}, x_j]$ and $t \in \left[\frac{i-1}{n}, \frac{i}{n}\right]$ then $\Phi(s,t) \in C_i$ then this shows that ϕ_{i+1} is an elementary deformation of ϕ_i. $\square$

2.6 Zeros

Suppose that $f : B(a,r) \to \mathbb{C}$ is holomorphic. By Taylor's theorem we know that

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$

and if $f \neq 0$ there is a least m such that $c_m \neq 0$. if $m > 0$ then $f(a) = 0$ we can write that $f(z) = (z-a)^m g(z)$ where g is holomorphic on $B(a,r)$ such that $g(a) \neq 0$. We say that f has a zero of order m at a.

Proposition. (Principle of isolated zeros) If $f : B(a,r) \to \mathbb{C}$ is holomorphic and $f \not\equiv= 0$ then there exists $0 < \rho < r$ such that

$$f(z) \neq 0 \quad \forall z \in B(a,\rho) \setminus \{a\} = B(a,\rho)^*.$$

Proof. If $f(a) \neq 0$ the result is immediate by continuity of f. If $f(a) = 0$ we said that $f(z) = (z-a)^m g(z)$ where $g(a) \neq 0$. By continuity of g, there exists a ρ such that g doesn't vanish on $B(a,\rho)$. Hence f doesn't vanish on $B(a,\rho)^*$. $\square$

Recall an Analysis definition

Definition. (Accumulation point) We say that a set $X \subseteq \mathbb{C}$ has an accumulation point at $a \in \mathbb{C}$ if for all $\varepsilon > 0$ we have that $B(a, \varepsilon)^* \cap X \neq \emptyset$.

Theorem. (Identity theorem) Let $U \subseteq \mathbb{C}$ be a domain and $f, g : U \to \mathbb{C}$ be holomorphic functions on U. Let $S = \{z \in U : f(z) = g(z)\}$. If S has an accumulation point in U then $f(z) = g(z)$ for all $z \in U$.

Proof. Let $h(z) = f(z) - g(z)$. Then h is holomorphic on U and vanishes on S. Suppose that w is an accumulation point of S. So it's a non-isolated zero of h. Hence there exists a $\varepsilon > 0$ such that $B(w, \varepsilon) \subseteq U$ and $h \equiv 0$ on the ball by the principle of isolated zeros. Let $z \in U$, so there is a path $\gamma : [0, 1] \to U$ such that $\gamma(0) = w$ and $\gamma(1) = z$. Let

$$\begin{aligned} I &= \{t \in [0, 1] \mid h^{(n)}(\gamma(t)) = 0 \ \forall n \geq 0\} \\ &= \{t \in [0, 1] \mid h \text{ vanishes on a small open neighbourhood of } \gamma(t)\} \end{aligned}$$

by Taylor's theorem. We know that I is non-empty since $0 \in I$. But the first line shows that I is open and the second line shows that I is closed. Hence I is clopen and non-empty, so $I = [0, 1]$ and $h(z) = 0$ so $h \equiv 0$ on U. □

Remark. Let's see some remarks from this theorem.

(i) If $f \not\equiv 0$ where f is holomorphic on U, then the zeros of f cannot accumulate in U. However they can accumulate at a point not in U. For example consider $U = \mathbb{C}^*$ and $f(z) = \sin\left(\frac{1}{z}\right)$. Then $f(z) = 0$ for $z = \frac{1}{k\pi}$ for $k \in \mathbb{N}$ so zeros do accumulate at 0.

(ii) Identity between entire functions holding on $\mathbb{R}$ also hold on $\mathbb{C}$, since we can rearrange them to zero and hence the function vanishes on the real axis so must vanish everywhere.

If U is a domain, we let $\overline{U}$ denote the closure of U. This is the smallest closed set containg U and the set of limit points of U. We write that $\partial U = \overline{U} \setminus U = \{z \in \mathbb{C} \mid \forall \varepsilon > 0, B(z, \varepsilon) \cap U \neq \emptyset\} \setminus U$.

Corollary. (Global maximum principle) Let $U \subseteq \mathbb{C}$ be a bounded d omain. Let $f : \overline{U} \to \mathbb{C}$ be continuous and holomorphic on U. Then $|f|$ achieves its maximum on ∂U.

Proof. If $U \subseteq B(0, N)$ then $\overline{U} \subseteq \overline{B(0, N)}$ so $\overline{U}$ is closed and bounded, so $|f| : \overline{U} \to \mathbb{R}$ does achieve it's maximum M on $\overline{U}$. Suppose that $|f(a)| = M$ with $a \in U$. Pick $\varepsilon > 0$ such that $B(a, \varepsilon) \subseteq U$. The local maximum principle implies that f is constant on $B(a, \varepsilon)$. Now f and a constant function agree on $B(a, \varepsilon) \subseteq U$ hence they agree everywhere on U. So f is constant on U hence f is constant on $\overline{U}$ by continuity, hence the maximum of $|f|$ also occurs on ∂U. □

Definition. (Analytic continuation) Let $U \subseteq U' \subseteq \mathbb{C}$ be domains and $f : U \to \mathbb{C}$ holomorphic. We say that a holomorphic function $h : U' \to \mathbb{C}$ such that $h\,|_U = f$ is an *analytic continuation* of f (to U').

Corollary. Analytic continuations are unique.

Proof. h and f agree on an open set hence by the identity theorem they agree on U'. □

Definition. (Natural boundary) If $f : U \to \mathbb{C}$ is holomorphic and admits no analytic continuation to any domain $U' \not\supseteq U$ we say that ∂U is the *natural boundary* of f.

Example. Let's see some functions and their natural boudaries.

(i) Let $f(z) = \sum_{n=0}^{\infty} z^n$ with radius of convergence 1 and define a holomorphic function on $\mathbb{D} = B(0,1)$. This has analytic continuation $\frac{1}{1-z}$ defined on $\mathbb{C} \setminus \{1\}$.

(ii) Let $f(z) = \sum_{n=0}^{\infty} z^{2^n}$ defines a holomorphic function on $\mathbb{D}$ with natural boundary $\partial\mathbb{D} = S^1$.

2.7 Singularities

Recall that if U is a domain $S \subseteq U$ finite and $f : U \to \mathbb{C}$ continuous and holomorphic on $U \setminus S$, then f is holomorphic on U by Morera's theorem. But what happens if f is not continuous at S?

Proposition. (Removal of singularities) Let U be a domain and $z_0 \in U$, $f : U \setminus \{z_0\} \to \mathbb{C}$ is holomorphic and f is bounded near z_0. Then there exists a $a \in \mathbb{C}$ such that $f(z) \to a$ as $z \to z_0$ and set

$$g(z) = \begin{cases} f(z) & z \in U \setminus \{z_0\} \\ a & z = z_0 \end{cases}$$

then g is holomorphic on U.

By hypothesis there exists a $\delta > 0$ and $M > 0$ such that $|f(z)| < M$ if $z \in B(z_0, S)^*$. Define $h : U \to \mathbb{C}$ by

$$h(z) = \begin{cases} (z - z_0)^2 f(z) & z \neq z_0 \\ 0 & z = z_0 \end{cases}$$

Then h is complex differentiable on U clearly for $z \neq z_0$. Near z_0 we have that

$$\left| \frac{h(z) - h(z_0)}{z - z_0} \right| \leq M|z - z_0|$$

so as $z \to z_0$

$$\lim_{z \to z_0} \frac{h(z) - h(z_0)}{z - z_0}$$

exists and is zero. Hence h is holomorpic so has a Taylor series convergent near z_0, hence $h(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$. Now $h(z_0) = 0$ and $h'(z_0) = 0$ so $a_0 = a_1 = 0$. Let

$$g(z) = \sum_{n=0}^{\infty} a_{n+2} (z - z_0)^n$$

which is also convergent on $B(z_0, \varepsilon)$. By construction g agrees with f on $B(z_0, \varepsilon)^*$, so extends to U by analytic continuation (as f). So indeed

$$\lim_{z \to z_0} f(z) = a_2$$

and we're done. $\square$

Proposition. Let U be a domain with $z_0 \in U$ and $f : U \setminus \{z_0\} \to \mathbb{C}$ be holomorphic. Suppose that $|f(z)| \to \infty$ as $z \to z_0$. Then there is a holomorphic function $g : U \to \mathbb{C}$ with $g(z_0) \neq 0$ and a $k \geq 1$ such that

$$f(z) = \frac{g(z)}{(z - z_0)^k}$$

for all $z \neq z_0$ with both g and k unique.

Proof. Pick $\delta > 0$ such that $|f(z)| \geq 1$ for all $z \in B(z_0, S06*$. Now let

$$h(z) = \begin{cases} \frac{1}{f(z)} & z \in B(z_0, \delta)^* \\ 0 & z = z_0 \end{cases}$$

so the previous removal of singularities proposition says that h is holomorphic on $B(z_0, \delta)$. From our discussion of zeros of holomorphic functions, there exsits a unique $k \geq 1$ such that h has a zero of order k at z_0. Then using the Taylor series for h we have that

$$h(z) = (z - z_0)^k \ell(z)$$

where $\ell : B(z_0, \delta) \to \mathbb{C}$ is holomorphic with $\ell(z_0) \neq 0$. ℓ is continuous so there exists some $0 < \varepsilon < \delta$ such that $\ell(z) \neq 0$ for all ε in $B(z_0), \varepsilon)$. Now let

$$g(z) = \frac{1}{\ell(z)}.$$

Then g is holomorphic in $B(z_0, \varepsilon)$ and $g(z) = (z - z_0)^k f(z)$ for all $z \in B(z_0, \varepsilon)^*$. Then the final expressions shows that g has an analytic continuation to U. $\square$

Definition. Let U be a domain with $z_0 \in U$ and $f : U \setminus \{z_0\} \to \mathbb{C}$ be holomorphic. We say that f has a *isolated signularity* at z_0 and that this singularity is

(i) *removable* if $|f|$ is bounded near z_0;

(ii) a *pole* if $|f| \to \infty$ near z_0 of order k if $f(z) = \frac{g(z)}{(z-z_0)^k}$ with g holomorphic and $g(z_0) \neq 0$;

(iii) *essential* otherwise.

We showed that in case (i) f has a limit at z_0 hence (iii) holds if and only if $|f|$ has no limit in $\mathbb{R} \cup \{\infty\}$. In case (ii) we saw that there is a unique $k \geq$ and holomorphic g on $B(a, r)$ such that $g(z_0) \neq 0$

$$f(z) = \frac{g(z)}{(z - a)^k}$$

and we say that the pole has order k.

Example. Let's see some examples.

(i) Take $f(z) = \frac{\sin(z)}{z}$ on $\mathbb{C}^*/$ We know that $\sin(z) = z - \frac{z^3}{3!} + \cdots$ so in fact

$$\frac{\sin(z)}{z} = 1 - \frac{z^2}{2!} + \cdots$$

(ii) $f(z) = \frac{1}{z^3}$ has an order 3 pole as $z = 0$.

(iii) $f(z) = \exp\left(\frac{1}{z}\right)$ defined on $\mathbb{C}^*$ at a essential singularity at $z = 0$. Recall that the conformal map exp sends the horizontal strip of width 2π to $\mathbb{C}^*$. But $z \to \frac{1}{z}$ sends $B(0, \varepsilon)^* \to \mathbb{C} \setminus \overline{B(0, \frac{1}{\varepsilon})}$ which contains such a 2π width strip. Hence the function $f(z) = \exp(1/z)$ is onto $\mathbb{C}^*$ in every punctured neighbourhood of the origin. Hence $|f|$ can have no limit at 0 so the singularity is essential.

Theorem. (Casorati-Weierstrass) If U is a domain and $f : U \setminus \{z_0\} \to \mathbb{C}$ is holomorphic has an essential singularity at $z_0 \in U$. Then for all $w \in \mathbb{C}$ there is some sequence $z_n \to z_0$ such that $f(z_n) \to w$.

Proof. Example Sheet 2. □

In fact something much stronger is true, which is the Great Picard's theorem.

Theorem. (Picard's Great theorem) If f is an analytic function with an essential singularity at a point z_0 then on any punctured neighbourhood of z_0, $f(z)$ takes all possible complex values, with at most one exception infinitely often.

Proof. Very involved, omitted.

Remark. This shows that $\mathbb{C}$ minus one point is very different analytically and topologically from $\mathbb{C}$ minus two or more points. This is covered in more detail in II Riemann Surfaces.

Remark. If $f : B(z_0, r)^* \to \mathbb{C}$ is holomorphic with a pole at z_0 then since $|f| \to \infty$ as $z \to z_0$, f defines a map from $B(z_0, r) \to \mathbb{C} \cup \{\infty\}$, defined by sending $f(z_0) = \infty$.

Definition. (Meromorphic) If U is a domain then $S \subset U$ is finite (discrete) and $f : U \setminus S \to \mathbb{C}$ is holomorphic with only poles and removable singulaties then we say that f is *meromorphic* on U.

Remark. When we say that S is *discrete* we mean that all the points are isolated, so there is a neighbourhood around every point disjoint from S. This definition means that a meromorphic function can have at most a countable number of poles since we can split $\mathbb{C}$ up into a union of compact subspaces. Each compact subspace can only contain a finite number of poles, otherwise there would be an accumulation point by Bolzano-Weiestrass which we can't have since S is discrete. Hence since the countable union of finite sets is countable, the total number of poles is countable.

Example. If $P, Q \in \mathbb{C}[z]$, then $\frac{P}{Q}$ is meromorphic on $\mathbb{C}$.

2.8 Laurent series

Theorem. (Laurent series) Let $A = \{z \in \mathbb{C} : r < |z - a| < R\}$ where $0 < r < R \leq \infty$ and let $f : A \to \mathbb{C}$ be holomorphic. Then

(i) f has a unique convergent power series expansion

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$$

on A.

(ii) If $r < \rho' \leq \rho < R$ the series converges uniformly on the annulus $\{\rho' \leq |x-a| \leq \rho\}$.

(iii) For $r < \rho < R$, we have that

$$c_n = \frac{1}{2\pi i} \int_{\partial B(a,\rho)} \frac{f(w)}{(w-a)^{n+1}} \, \mathrm{d}w.$$

Corollary. If $r = 0$ then $A = B(a,R)^*$ and f has an isolated singularity at a. Then

(i) a is removable if and only if $c_n = 0$ for all $n < 0$

(ii) a is a pole order k if and only if $c_k \neq 0$ and $c_n = 0$ for all $n < -k$.

(iii) a is essential if and only if $c_n \neq 0$ for infinitely many negative n.

Proof. (i) and (ii) Immediate from our previous characteristions and then (iii) is true by exclusion. □

Remark. In Taylor's theorem, $f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$ if f is holomorphic on $B(a,R)$ and

$$c_n = \frac{f^{(n)}(a)}{n!}$$

but in Laurent's theorem where f is holomorphic on $B(a,R)^*$ there is no general formula for these coefficients.

Proof. Pick ρ_1, ρ_2 such that $r < \rho_1 < |z-a| < \rho_2 < R$ where we fixed $z \in A$. Let γ_1, γ_2 be piecewise-C^1 curves formed by cutting from the ρ_2 circle down to the ρ_1 circle forming two half annuli. Let γ_1 be the boundary of the half annulus containing z traversed anticlockwise and γ_2 be the boundary of the other half annulus traversed clockwise. Applying the Cauchy integral formula for γ_1, we get that

$$f(z) = \frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w-z} \, \mathrm{d}w.$$

Applying Cauchy's theorem for γ_2 we get that

$$0 = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w-z} \, \mathrm{d}w.$$

Adding these, note that the line segments of γ_1 and γ_2 appear with opposite orientation. Hence we deduce that

$$f(z) = \frac{1}{2\pi i} \left(\int_{\partial B(a,\rho_2)} \frac{f(w)}{w-z} \, \mathrm{d}w - \int_{\partial B(a,\rho_1)} \frac{f(w)}{w-z} \, \mathrm{d}w \right).$$

As in the proof of Taylor's theorem we say that

$$\frac{1}{w-z} = \frac{1}{w-a} \frac{1}{1 - \frac{z-a}{w-a}} = \sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}}$$

which coverges uniformly on $|w-a| = \rho_2$ since $|z-a| < \rho_2$. We'll use this for the first integral. For the second integral, we use that

$$-\frac{1}{w-z} = \frac{1}{z-a}\frac{1}{1-\frac{w-a}{z-a}} = \sum_{m=1}^{\infty} \frac{(w-a)^{m-1}}{(z-a)^m}$$

which coverges uniformly on $|w-a| = \rho_1$ since $|z-a| > \rho_1$. Substitute this into the second integral. In both cases uniform convergence lets us swap the integral and summation. Thus we get that

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial B(a,\rho_2)} \frac{f(w)}{(w-a)^{n+1}} \, \mathrm{d}w \right) (z-a)^n + \sum_{m=1}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial B(a,\rho_1)} \frac{f(w)}{(w-a)^{-m+1}} \, \mathrm{d}w \right) (z-a)^{-m}$$

Now relabel $n = -m$ in the second series and replace both contours $\partial B(a,\rho_2)$ and $\partial B(a,\rho_1)$ by a common contour $|w-a| = \rho$ using homotopy invariance of the integral. Thus we get the Laurent series.

It remains to show uniqueness. If we could write

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n = \sum_{m=-\infty}^{\infty} b_m (z-a)^m$$

then

$$\begin{aligned} 2\pi i c_k = \int_{\partial B(a,\rho)} \frac{f(w)}{(w-a)^{n+1}} \, \mathrm{d}w &= \int_{\partial B(a,\rho)} \sum_{m=-\infty}^{\infty} b_m (w-a)^{m-k-1} \, \mathrm{d}w \\ &= \sum_{m=-\infty}^{\infty} b_m \int_{\partial B(a,\rho)} (w-a)^{m-k-1} \, \mathrm{d}w = 2\pi i b_k \end{aligned}$$

Thus we have unique Laurent series and the rest of the theorem follows. □

Definition. (Laurent series) For $a \in \mathbb{C}$ a *Laurent series* centred at a is a sum

$$\sum_{n=-\infty}^{\infty} c_n (z-a)^n.$$

We say the series converges at some $z \in \mathbb{C}$ if $\sum_{n=0}^{\infty} c_n (z-a)^n$ and the sum $\sum_{n=-\infty}^{-1} c_n (z-a) = \sum_{n=1}^{\infty} c_n (z-a)^{-n}$ converges.

From our results on power series, there exists a radius of convergence $R \geq 0$ such that $\sum_{n=0}^{\infty} c_n (z-a)^n$ converges absolutely on $\{|z-a| < R\}$ and coverges uniformly on $\{|z-a| \leq \rho\}$ for $\rho < R$.

Similarly if we set $u = \frac{1}{z-a}$ then

$$\sum_{n=1}^{\infty} c_{-n} (z-a)^{-n} = \sum_{n=1}^{\infty} c_{-n} u^n$$

has a radius of convergence R' such that it converges absolutely if $|u| < R'$ i.e. if $|z-a| > \frac{1}{R'} = r$. If $r < R$, the Laurent series converges absolutely on the annulus $A = \{r < |z-a| < R\}$ and

converges uniformly on $\{\rho \leq |z-a| \leq \rho'\}$ for $r < \rho \leq \rho' < R$. So Laurent's theorem says that if f is holomorphic on an annulus A it admits a unique such Laurent series centred at the centre of the annulus. In our proof, we explicitly constructed power series f_{out} and f_{in} where f_{in} was holomorphic on $|z-a| < R$ and f_{out} was holomorphic on $|z-a| > r$.

Definition. (Principal part) If

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$$

on an annulus A then the sum

$$\sum_{n=-\infty}^{-1} c_n (z-a)^n$$

is called the *principal part* of f, written as f_{prin}.

Remark. If f is holomorphic on $B(a,R)^*$ (so $r=0$), then f_{prin} defines a holomorphic function on $\mathbb{C} \setminus \{a\}$, whcih moreover vanishes at infinity since the power series in $u = \frac{1}{z-a}$ has no constant term.

Again note that (when $r=0$) $f - f_{\text{prin}}$ is holomorphic near a.

Example. Consider $\csc(x) = \frac{1}{\sin(z)}$ which is holomorphic everywhere except at $z = k\pi$ for $k \in \mathbb{Z}$. So it has a Laurent series on $\{0 < |z| < \pi\}$. Now we know that

$$\sin z = z\left(1 - \frac{z^2}{6} + O(z^4)\right)$$

so

$$\csc z = \frac{1}{z}\left(1 - \frac{z^2}{6} + O(z^4)\right)^{-1} = \frac{1}{z}\left(1 + \frac{z^2}{6} + O(z^4)\right)$$

hence $c_{-1} = 1$ and csc has a simple pole at 0 and $c_k = 0$ for all $k < -1$.

Example. We claim that if $z \in \mathbb{C} \setminus \mathbb{Z}$, we have that

$$\sum_{n \in Z} \frac{1}{(z-n)^2} = \frac{\pi^2}{\sin^2(\pi z)}.$$

For $z \in \mathbb{C} \setminus \mathbb{Z}$, set $f(z) = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}$. If $w \in \mathbb{C} \setminus \mathbb{Z}$ we can pick $r > 0$ such that $|w-n| > 2r$ for all $n \in \mathbb{Z}$. Then $\forall z \in B(\omega, r)$, we have that $|z-n| \geq \max\{r, n - |\omega| - r\}$. Hence

$$\frac{1}{|z-n|^2} = \min\left\{\frac{1}{r^2}, \frac{1}{(n-|\omega|-r)^2}\right\} = M_n.$$

We have that $\sum_{n \geq N} M_n$ converges by comparison to $\sum \frac{1}{n^2}$ so by the Weiestrass M-test we have that f converges uniformly on $B(\omega, r)$. So f is holomorphic on $\mathbb{C} \setminus \mathbb{Z}$. Note that

(i) $f(z+1) = f(z)$;

(ii) $\forall n \in \mathbb{Z}$, f has a double pole.

Let

$$g(z) = \left(\frac{\pi}{\sin \pi z}\right)^2$$

hence g is also $\mathbb{Z}$-periodic, so $g(z) = g(z+1)$. So from our power series for sin and csc g also has double poles at $n \in \mathbb{Z}$. At $z = n$, $f_{\text{prin}} = \frac{1}{(z-n)^2} = g_{\text{princ}}$ by checking that $\lim_{z\to 0} z^2 g(z) = 1$ and using periodicity. So $f(z) - g(z)$ is holomorphic on $\mathbb{C} \setminus \mathbb{Z}$ and has trivial principal part at each $n \in \mathbb{Z}$. Hence $h(z) = f(z) - g(z)$ is entire. By Liouville it's enough to show that h is bounded and $h \to 0$ near infinity.

By periodicity we can just consider h on $-\frac{1}{2} \leq \text{re}z \leq \frac{1}{2}$. If $z = x + yi$ with $|x| \leq \frac{1}{2}$, $y > 0$ then

$$|g(z)| = \frac{4\pi^2}{|e^{\pi y} - e^{-\pi y}|^2} \to 0$$

as $y \to \infty$, and

$$|f(z)| \leq \sum_{n\in\mathbb{Z}} \frac{1}{|x + yi - n|^2} \leq \frac{1}{y^2} + \sum_{n=1}^{\infty} 2\left(\frac{1}{(n - \frac{1}{2})^2 + y^2)}\right) \to 0$$

as $y \to \infty$. So there exists a $Y > 0$ such that $|f|, |g|$ are bounded by 1 outside the box hence they are bounded in the box by continuity. So we've proved the identity.

3 Residues

3.1 Winding numbers

We've seen that the integral "detects" the number of times a curve "winds" around a singularity giving a non-zero value sometimes when we integrate around a singularity of a function, for example $\frac{1}{z}$.

Lemma. Let $\gamma : [a, b] \to \mathbb{C}\setminus\{w\}$ be path such that $w \notin \text{im}(\gamma)$. Then there are continuous functions

$$r : [a, b] \to \mathbb{R}$$
$$\theta : [a, b] \to \mathbb{R}$$

such that $\gamma(t) = w + r(t)e^{i\theta(t)}$.

Proof. By translating *wlog* $w = 0$. We set $r(t) = |\gamma(t)|$. Replacing γ by $\frac{\gamma}{r}$ we can suppose that $|\gamma(t)| = 1$ for all t.

If we had that $\gamma(t) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}$ we could set $\theta = \text{Arg}(\gamma(t)) \in (-\pi, \pi$. More generally if $\gamma(t) \subseteq \mathbb{C} \setminus \mathbb{R}_{\leq 0}(e^{i\alpha})$ then there exists a branch of the logarithm on the image of γ so we can define θ for example by $\text{Arg}\left(\frac{z}{e^{i\alpha}}\right)$. We have that γ is continuous on $[a, b]$ hence it is uniformly continous, so there exists a $\delta > 0$ such that for $|s - t| < \delta$ we have $|\gamma(s) - \gamma(t)| < 2$. So $\gamma(s)$ and $\gamma(t)$ are antipodean. Write $a = a_0 < a_1 < \cdots < a_N = b$ such that $|a_{i-1} - a_i| < \delta$ and $\gamma|_{a_{i-1},a_i}$ has an image in a slit plane. So for all n there exists a $\theta_n : [a_{n-1}, a_n] \to \mathbb{R}$ such that $\gamma(t) = e^{i\theta_n(t)}$ on $[a_{n-1}, a_n]$ for $n = 1, 2, \ldots, N$. Note that $\theta_{n-1}(a_{n-1}) = \theta_n(a_{n-1}) + b_n$ for some $b_n \in 2\pi\mathbb{Z}$ since values of the argument is fixed. Hence we can redefine θ so they join up so θ is continuous. $\square$

Remark. In our construction, $\theta(t)$ is not unique. if $\gamma(t) = w + r(t)e^{i\theta(t)} = w + r(t)e^{i\tilde{\theta}(t)}$ both continuous, hence $\theta - \tilde{\theta} : [a, b] \to 2\pi\mathbb{Z} \subseteq \mathbb{R}$. So $\theta - \tilde{\theta}$ is constant. Hence θ is unique up to an addition of $2\pi k$ for some fixed k.

Definition. (Winding number) Now suppose that $\gamma : [a, b] \to \mathbb{C} \setminus \{w\}$ is a continuous *closed* path, so $\gamma(a) = \gamma(b)$. Then the *winding number* $I(\gamma; w)$ of γ about w is

$$\frac{\theta(b) - \theta(a)}{2\pi}$$

where θ is defined above.

Remark. Our previous remark shows that $I(\gamma, w)$ is well-defined and in $\mathbb{Z}$.

Lemma. If $\gamma : [a, b] \to \mathbb{C} \setminus \{w\}$ is a piecewise-C^1 closed path,

$$I(\gamma; w) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z - w} \, \mathrm{d}z.$$

Proof. When γ is a piecewise-C^1 curve both $r(t)$ and $\theta(t)$ are also piecewise-C^1. Then

$$\int_\gamma \frac{1}{z - w} \, \mathrm{d}z = \int_a^b \frac{\gamma'(t)}{\gamma(t) - w} \, \mathrm{d}t = \int_a^b \frac{r'(t)}{r(t)} + i\theta'(t) \, \mathrm{d}t$$

if $\gamma(t) = w + r(t)e^{i\theta(t)}$. This gives

$$= \log r(t) + i\theta(t)|_a^b = i(\theta(b) - \theta(a)) = 2\pi i I(\gamma, w). \quad \square$$

In general if $\gamma(t)$ is a loop, we can consider $I(\gamma, w)$ as a function on points $w \notin \mathrm{im}(\gamma)$. In general, $\mathbb{C} \setminus \mathrm{im}(\gamma)$ may have many path components. Note there is exactly one path component that is unbounded since $\mathrm{im}(\gamma)$ is a closed and bounded set so we can always contain it in some $B(0, R)$ for R sufficiently large. Then the path component containing $\mathbb{C} \setminus B(0, R)$ is unbounded.

Proposition. Let γ be a closed curve and $w \notin \mathrm{im}\,\gamma$. Then
(i) $I(\gamma, w)$ is constant on path-components of $\mathbb{C} \setminus \mathrm{im}\,\gamma$.
(ii) On the unbounded component $I(\gamma, w) = 0$.

Proof. The expression $I(\gamma, w) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z-w} \mathrm{d}z$ is continuous in w. So $I(\gamma, w) : \mathbb{C} \setminus \mathrm{im}\,\gamma \to \mathbb{Z}$ is continuous hence it's constant on path so constant on path components.

For the second part, we know that $\mathrm{im}\,\gamma \subseteq B(0, R)$ for some $R > 0$. Pick w such that $|w| > 2R$. Then

$$\begin{aligned} |I(\gamma, w)| &= \frac{1}{2\pi} \left| \int_\gamma \frac{\mathrm{d}}{z - w} \right| \\ &\leq \frac{\mathrm{length}(\gamma)}{2\pi} \frac{1}{R}. \end{aligned}$$

As $R \to \infty$ the RHS < 1 and $I(\gamma, w) \in \mathbb{Z}$ so it must be zero. $\square$

Remark. Consider Example Sheet 3 for a related viewpoint.

3.2 Cauchy's residue theorem

Let's go back to considering holomorphic functions $f : B(a,r)^* \to \mathbb{C}$ with an isolated singularity at a and hence a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n.$$

Definition. (Residue) The value c_{-1} is called the *residue* of f at a, written as $\mathrm{Res}_f(a)$.

So if $0 < \rho < r$, we have that

$$\int_{\partial B(a,\rho)} f(z)\ \mathrm{d}z = 2\pi c_{-1} = 2\pi i \,\mathrm{Res}_f(a).$$

We are now ready to relate integrals and winding numbers together in a theorem.

Theorem. (Cauchy's residue theorem) Let U be a simply connected domain and let $\{z_1, \dots, z_k\} \subseteq U$ and $f : U \setminus \{z_1, \dots, z_k\} \to \mathbb{C}$ be holomorphic. Then let $\gamma : [a,b] \to U \setminus \{z_1, \dots, z_k\}$ be a piecewise-C^1 closed path. Then

$$\frac{1}{2\pi i} \int_\gamma f(z)\mathrm{d}z = \sum_{j=1}^{k} I(\gamma, z_j)\,\mathrm{Res}_f(z_j).$$

Proof. At z_j, f has a Laurent series $\sum_{n=-\infty}^{\infty} c_n^{(j)}(z-z_j)^n$ convergent in some $B(z_j,\rho_j)^* \subseteq U$. Let

$$g_j(z) = \sum_{n=-\infty}^{-1} c_n^{(j)}(z-z_j)^n$$

be the principal part of this series. This is holomorphic on $\mathbb{C} \setminus \{z_j\}$ so in particular on $U \setminus \{z_j\}$. Consider $f - g_1 - \cdots - g_k$ which is now holomorphic on $U \setminus \{z_1, \dots, z_k\}$, with removable singularities at each z_i so this defines a holomorphic function on the whole of U. Now we can apply Cauchy's theorem for a simply connected domain, so

$$\int_\gamma (f - g_1 - \cdots - g_k) = 0$$

so rearranging we get that

$$\int_\gamma f = \sum_{j=1}^{k} \int_\gamma g_j(z)\ \mathrm{d}z = \sum_{j=1}^{k} c^{(j)} 2\pi i I(\gamma, z_j)$$

and since $\mathrm{Res}_f(z_j) = \mathbf{??} g_j(z_j)$ we see

$$\int_\gamma f = 2\pi i \sum_{j=1}^{k} \mathrm{Res}_f(z_j) I(\gamma, z_j).$$

giving the result. $\square$

Remark. This is a generalisation of Cauchy's theorem (f only having removable singularities), but also a generalisation of the Cauchy integral formula by applying the residue theorem to the function

$$\hat{f}(z) = \frac{f(z)}{z-w}$$

and the contour $\gamma(t) = \partial B(w, \rho)$.

Remark. We say a piecewise-C^1 closed curve γ bounds a domain D in U if $I(\gamma, w) = \mathbf{1}_{\{w \in D\}}$ for $w \in U \setminus \operatorname{im} \gamma$. This is always true if γ is simple and U is simply connected. This is the special case of the Jordan curve theorem.

We have a special case of the residue theorem which is more useful for the direct computation of integrals.

Corollary. If f is meromorphic on U and has no poles on γ and γ bounds a domain D in U then

$$\int_\gamma f = 2\pi i \sum_{w \text{ pole in } D} \operatorname{Res}_f(w)$$

We've already proved there's only finitely many poles in D so this holds from the residue theorem.

Lemma. Let $f : U \to \mathbb{C}$ be meromorphic with a pole at $a \in U$.

(i) If the pole is simple then

$$\operatorname{Res}_f(a) = \lim_{z \to a} (z-a) f(z).$$

(ii) If $f(z) = \frac{g(z)}{h(z)}$ where h, g are holomorphic at a, $g(a) \neq 0$ and h has a simple zero at a then

$$\operatorname{Res}_f(a) = \frac{g(a)}{h'(a)}.$$

(iii) If $f(z) = \frac{g(z)}{(z-a)^k}$ with g holomorphic near a then

$$\operatorname{Res}_f(a) = \frac{g^{(k-1)}(a)}{(k-1)!}.$$

Proof.

(i) $f(z) = \frac{c_{-1}}{z-a} + c_0 + c_1(z-a) + \dots$. So clearly, $c_{-1} = \lim_{z \to a}(z-a)f(z)$.

(ii) Using (i), $\operatorname{Res}_f(a) = \lim_{z \to a}(z-a)\frac{g(z)}{h(z)}$ since h having a simple zero implies that $\frac{1}{h}$ has a simple pole. Hence we have that

$$\begin{aligned} \operatorname{Res}_f(a) &= g(a) \lim_{z \to a} \frac{z-a}{h(z)-h(a)} \\ &= \frac{g(a)}{h'(a)}. \end{aligned}$$

(iii) We want the coefficient of $(z-a)^{k-1}$ in Taylor series for holomorphic function g. Then this is just part of Taylor's theorem. $\square$

3.3 Applications of residues for integration

Example. Consider the function

$$f(z) = \frac{1}{1+z^4}$$

This is meromorphic on $\mathbb{C}$ with poles at the 4th roots of unity. Consider a contour γ_R which runs along a long stetch of the real axis from $-r$ to r and the along a semicircle of radius r. This encloses two poles of f at $e^{i\pi/4}$ and $e^{3i\pi/4}$. So γ_R encloses two simple poles of F called p_1 and p_2 and its winding number about each is obviously 1. Then

$$\int_{\gamma_R} f(z)\ \mathrm{d}z = \int_{-R}^{R} \frac{1}{1+x^4}\ \mathrm{d}x + \int_0^{\pi} \frac{iRe^{i\theta}}{1+R^4e^{4i\theta}}\ \mathrm{d}\theta.$$

We can see our second integral is $O(1/R^3)$ so goes to zero as $R \to \infty$. Apply the residue theorem so in the limit at $R \to \infty$ we have that

$$\begin{aligned} 2\int_0^{\infty} \frac{1}{1+x^4}\ \mathrm{d}x &= 2\pi i(\mathrm{Res}_f(p_1) + \mathrm{Res}_f(p_2) \\ &= -2\pi i(\frac{1}{4}(p_1+p_2)). \end{aligned}$$

Hence we have that

$$\int_0^{\infty} \frac{1}{1+x^4}\ \mathrm{d}x = \frac{\pi}{2\sqrt{2}}.$$

Example. Now we want to evaulate the integral

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x+x^2}\ \mathrm{d}x.$$

At $x = iR$ $\cos z \approx e^R$ so we can't easily bound the integrand on the same kind of semicircular contour. Let

$$f(z) = \frac{e^{iz}}{1+z+z^2}$$

then the integrand is the real part of f which has poles only at the zeros of $1+z+z^2$. Now we can use the contour γ_R as before. On the semicricle,

$$\left|\int_0^{\pi} f(Re^{i\theta})iRe^{i\theta}\ \mathrm{d}\theta\right| = \int_0^{2\pi} \frac{Re^{-R\sin\theta}}{|R^2e^{2i\theta}+Re^{i\theta}+1|}\ \mathrm{d}\theta.$$

This is $O(R^{-1})$ hence it vanishes as $R \to \infty$. So the arc section of the contour integral goes to zero. Hence

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\cos x}{1+x+x^2} &= \mathrm{Re}\int_{-\infty}^{\infty} f(z)\ \mathrm{d}z \\ &= \mathrm{Re}\lim_{R\to\infty}\int_{\gamma_R} f(z)\ \mathrm{d}z \\ &= \mathrm{Re}(2\pi i\,\mathrm{Res}_f(\omega)) \\ &= \frac{2\pi}{\sqrt{3}}e^{-\sqrt{3}/2}\cos\left(\frac{1}{2}\right) \end{aligned}$$

Lemma. Let $f : B(a,r)^* \to \mathbb{C}$ be holomorphic with a simple pole at a. For $0 < \varepsilon < r$, consider $\gamma_\varepsilon : [\alpha, \beta] \to \mathbb{C}$ sending $t \to a + \varepsilon e^{it}$. Then

$$\lim_{\varepsilon \to 0} \int_{\gamma_\varepsilon} f(z)\ \mathrm{d}z = (\beta - \alpha) i \operatorname{Res}_f(a).$$

Proof. Since f has a simple pole, we can write that $f(z) = \frac{c}{z-a} + g(z)$ where g is holomorphic. g is continuous and locally bounded, so

$$\left| \int_{\gamma_\varepsilon} g \right| \leq (\beta - \alpha)\varepsilon \sup_{z \in \gamma_\varepsilon} |g(z)|$$

which goes to zero as $\varepsilon \to 0$. But

$$\lim_{\varepsilon \to 0} \int_{\gamma_\varepsilon} \frac{c}{z-a}\ \mathrm{d}z = c \lim_{\varepsilon \to 0} \int_\alpha^\beta \frac{1}{\varepsilon e^{it}} i\varepsilon e^{it}\ \mathrm{d}t = (\beta - \alpha) i \operatorname{Res}_f(a). \quad \square$$

Lemma. (Jordan's Lemma) Suppose that f is holomorphic on $\{|z| \geq r\}$ and that $zf(z)$ is bounded for large $|z|$. Then for all $\alpha > 0$,

$$\int_{\gamma_R} f(z) e^{i\alpha z}\ \mathrm{d}z \to 0$$

as $R \to \infty$ where $\gamma_R : [0, \pi] \to \mathbb{C}$ sends $t \to Re^{it}$.

Proof. We know that $|f(z)| \leq \frac{M}{|z|}$ for large $|z|$ and some $M > 0$. We also have that

$$|e^{i\alpha z}| = e^{-R\alpha \sin t}$$

on our contour γ_R. We note that on the interval $(0, \frac{\pi}{2}]$ the function $\frac{\sin\theta}{\theta}$ is decreasing since it's derviative is negative. So on $t \in [0, \frac{\pi}{2}]$ we have that

$$\sin t \geq \frac{2t}{\pi}.$$

which gives the bound

$$|e^{i\alpha z}| = e^{-R\alpha \sin t} \leq \begin{cases} e^{-Ra2t/\pi} & 0 \leq t \leq \frac{\pi}{2} \\ e^{-Ra2t'/\pi} & 0 \leq t' = \pi - t \leq \frac{\pi}{2} \end{cases}$$

Hence we can bound the first circle quarter arc as

$$\begin{aligned} \left| \int_0^{\pi/2} e^{iR\alpha e^{it}} f(Re^{it}) Re^{it}\ \mathrm{d}t \right| &\leq \int_0^{2\pi} e^{-2\alpha Rt/\pi} M\ \mathrm{d}t \\ &= \frac{M}{2\alpha R}(1 - e^{-\alpha R}) \to 0 \end{aligned}$$

as $R \to \infty$. The other quarter circle goes to zero too, hence the whole integral goes to zero. $\square$

Example. We want to show that

$$\int_0^\infty \frac{\sin x}{x}\ \mathrm{d}x = \frac{\pi}{2}.$$

If we get $\hat{f}(z) = \frac{\sin x}{x}$, the singularity at 0 is removable but it is $O(e^R/R)$ on γ_R.

If we instead let $f(z) = \frac{e^{iz}}{z}$ then this has a pole at 0 so we can't use a contour which runs through zero. Hence we'll choose the contour $\gamma_{R,\varepsilon}$ which is a closed half annulus of radii $\varepsilon < R$ hence the winding number around the pole is zero, so the integral is 0 for any $R, \varepsilon > 0$. If we take $R \to \infty, \varepsilon \to 0$ we should recover the contour we wanted to integrate across.

Using Jordan's lemam on the R-semicricle γ_R with $\alpha = 1$ and $\frac{1}{z}$ as the fuction we know that

$$\int_{\gamma_R} f(z)\ \mathrm{d}z \to 0$$

as $R \to \infty$. For the ε-semicircle, γ_ε using our first lemma we get a contribution of the negative residue $-i\pi$ since the orientation is reversed. Rearranging and using the fact that the function is even we get the desired result.

Example. Now we want to show that

$$\int_{-\infty}^\infty \frac{e^{ax}}{\cosh x}\ \mathrm{d}x = \pi \sec \frac{\pi a}{2}.$$

We have poles at $\cosh z = 0$ which corresponds to $z = \frac{2n+1}{2} i\pi$. This leads us to avoid semicircular contours t oavoid infinitely many poles in the limit. But $\cosh(x + i\pi) = -\cosh x$, hence we are lead to consider the rectangular contour going through $0, \pi i$ stretching out to R and $-R$ on the real axis. Now we only enclose a singular singularity, namely $\rho = \frac{i\pi}{2}$.

Along the right hand verticle segment,

$$\begin{aligned}\left|\int_\gamma f(z)\ \mathbf{d}z\right| &= \left|\int_0^\pi \frac{e^{a(R+iy)}}{\cosh(R+iy)} i\ \mathrm{d}y\right| \\ &\leq \int_0^\pi \frac{2e^{aR}}{|e^R - e^{-R}}\ \mathrm{d}y\end{aligned}$$

which goes toe zero as $R \to \infty$ since $a < 1$. On the left hand segment we need $a > -1$ to control the integral. Thus applying residue theorem we get that

$$\int_{-\infty}^\infty \frac{e^{ax}}{\cosh x} = \pi \sec\left(\frac{\pi a}{2}\right)$$

which gives the required result given that $a \in (-1, 1)$.

Example. We want to show that

$$\int_0^\infty \frac{\log x}{1 + x^2}\ \mathrm{d}x = 0.$$

We need a cut plane to define the logarithm. Cut along $\mathbb{R}_{\leq 0}$. We use a similar contour $\gamma_{R,\varepsilon}$ formed from the closed half of an annulus with radii ε and R. On the large semicircle the integrand is

$O\left(\frac{R\log R}{R^2}\right) \to 0$ as $R \to \infty$. On the small semicircle the integrand is order $O(\varepsilon \log \varepsilon) \to 0$ as $\varepsilon \to 0$. Along $\mathbb{R}_{\leq 0}$, $\log x = \log|x| + i\pi$. Applying the Cauchy residue thereom we get that

$$\int_{\gamma_{R,\varepsilon}} f(z)\ \mathrm{d}z = 2\pi i \operatorname{Res}_f(i)$$

where $f(x) = \frac{\log x}{1+x^2}$. Now assembling we have that

$$\int_0^\infty \frac{\log x}{1+x^2}\ \mathrm{d}x + \int_{-\infty}^\infty \frac{\log|x| + i\pi}{1+x^2}(-\mathbf{x}) = 2\pi \operatorname{Res}_f(i)$$

hence

$$2\int_0^\infty \frac{\log x}{1+x^2}\ \mathrm{d}x = 0$$

by comparing imaginary and real parts. □

Remark. We sometimes want a keyhole contour, where we use a slice taken out of an annulus to go around a branch cut but get arbitrarily close to it.

3.4 The argument principle

Definition. (Order of poles and zeros) Let $f : B(a,r)^* \to \mathbb{C}$ be meromorphic. Then we know that we can write

$$f(z) = (z-a)^k g(z)$$

with $g : B(a,r) \to \mathbb{C}$ holomorphic with $g(a) \neq 0$. Then we say that f has a zero of order k if $k > 0$. If $k < 0$ we say that f has a pole of order $|k|$. We write that $\operatorname{ord}_f(a) > 0$.

Lemma. Suppose that f is meromorphic on a domain U with a zero/pole of order k at $a \in U$. Then $\frac{f'(z)}{f(z)}$ has a simple pole at a of residue $\operatorname{ord}_f(a)$.

Proof. If $f(z) = (z-a)^k g(z)$ with $g(a) \neq 0$ then $f'(z) = k(z-a)^{k-1}g(z) + (z-a)^k g'(z)$, so

$$\frac{f'(z)}{f(z)} = \frac{k}{z-a} + \frac{g'(z)}{g(z)}.$$

The second term is holomorphic near a hence we have a simple pole with residue k. □

Definition. (Logarithmic derivative) The function $\frac{f'(z)}{f(z)}$ is called the *logarithmic derivative* of f.

Remark. If U is simply connected, and $0 \notin U$ we can define a branch of the logarithm on U.

Indeed this is true since $\frac{1}{z}$ is holomorphic on U, so $\int_\gamma \frac{1}{z}\ \mathrm{d}z = 0$ for all closed γ by simple connected Cauchy. Hence by the converse of the fundemental theorem of calculus there exists some antiderivative of $\frac{1}{z}$ which *wlog* is the logarithm. So if f is non-vanishing then $\log f(z)$ makes sense on U which has derivative $\frac{f'}{f}$.

Theorem. (The Argument Principle) Let U be a simple connected domain. Suppose that f is a meromorphic function on U with zeros $\{z_1, \dots, z_k\}$ and poles $\{w_1, \dots, w_\ell\}$. let $\gamma : [a, b] \to U$ be a closed piecewise-C^1 path which doesn't go through any zeros or poles. Then

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} \, \mathrm{d}z = \sum_{i=1}^{k} \mathrm{ord}_f(z_i) I(\gamma, z_i) - \sum_{j=1}^{\ell} \mathrm{ord}_f(w_j) I(\gamma, z_i).$$

Proof. The function $\frac{f'}{f}$ has a singularity at $z \in U$ if and only if z is a zero or pole of f. Now just apply the Cauchy residue theorem. □

Suppose γ bounds a domain D in U, so $I(\gamma, w) = \mathbb{1}(_{\{w \in D\}}$. Recall that if γ is simple and U is simply connected then is always true by the Jordan curve theorem. Then we get that

$$\frac{1}{2\pi i} \int_\gamma \frac{f'}{f} \, \mathrm{d}z = \text{Number of zeros in D} - \text{Number of poles in D}$$

each counted according to their multiplicity.

Remark. Let γ has domain $[a, b]$ and $\Gamma : [a, b] \to \mathbb{C}$ be $f \circ \gamma$. The zeros of f do not lie on γ, so Γ has image in $\mathbb{C}^*$. We know that γ is closed hence Γ is closed, so

$$\frac{1}{2\pi i} \int_\Gamma \frac{f'(z)}{f(z)} \, \mathrm{d}z = \frac{1}{2\pi} \int_\Gamma \frac{1}{w} \, \mathrm{d}w = I(\Gamma, 0).$$

Theorem. (Rouché's Theorem) Let $U \subseteq \mathbb{C}$ be a domain and let γ be a closed curve which bounds a domain D in U. Let f, g be holomorphic on U and suppose that $|f| > |g|$ on γ. Then f and $f + g$ have the same number of zeros in D when counted with multiplicity.

Proof. If $|f| > |g|$ on γ then in particular neither f or $f + g$ can vanish on γ. Let's look at

$$h(z) = \frac{f(z) + g(z)}{f(z)} = 1 + \frac{g(z)}{h(z)}.$$

Note that for all $z \in \gamma$, $h(z) \in B(1, 1)$ so $h \circ \gamma$ has image in $\mathrm{Re}(z) > 0$ hence $I(h \circ \gamma, 0) = 0$. The the argument principle says that the number of zeros of h in S agree, counted with multiplicity. This is exactly what we want since zeros of h correspond to zeros of $f + g$ and poles of h correspond to zeros of f. □

Example. We want to show that the polynomial $z^6 + 6z + 3$ has three roots (counted with multiplicity in the annulus $\{1 < |z| < 2\}$.

On $|z| = 2$ we have that $|z^4| = 16 > 6|z| + 3 > |6z + 3|$ so our polynomial has the same number of roots in $\{|z| < 2\}$ as z^4 i.e. 4 with multiplicity. On $|z| = 1$, $6|z| = 6 > |z^4 + 3|$ so there is only one root in $|z| < 1$ hence there are 3 roots in the annulus by Fundemental theorem of calculus.

Remark. FTA tells us we have roots: Roché tells us how to locate them.

Example. Let $p(z) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n \neq 0$ and $|a_{n-1}| > 1 + |a_{n-2}| + \cdots + |a_1| + |a_0|$, then p is irreducible in $\mathbb{Z}[x]$.

Let $f = a_{n-1}z^{n-1}$ and $g(z) = p(x) - f(x)$. Our hypothesis is then that $|f| > |g|$ on $|z| = 1$ So f and $f + g$ have $n - 1$ roots in $\{|z| < 1\}$. So p has exactly one root outside of the disc. Suppose that $p(x) = q(x)r(x)$ with q, r non-constant. So at least one of q and r has all roots inside the open unit disc. But $a_0 \neq 0$ so neither q, r have vanishing constant terms. But they have integer coefficients which is a contradiciton since the product of roots must be a non-zero integer. So p is irreducible.

Example. We claim that

$$\int_{|z|=2} \frac{1}{z^5 + 12z - 5} \mathrm{d}z = 0.$$

On the contour, $|z^5| = 32 > 29 \geq |12z - 5|$ so all of the zeros of $z^5 + 12z - 5$ lies inside $B(0, 2)$. Note that if we took the contour out to infinity, we have that

$$\int_{|z|=R} \frac{1}{z^5 + 12z - 5} \mathrm{d}z = O(R^{-4}) \to 0$$

as $R \to \infty$. Take the contour which is an annulus with radii 2 and R split down the positive imaginary axis with lines split ε apart. Then since the bound domain is simply connected (homotopic to the constant loop), the integral around it is zero since it contains no poles. As $\varepsilon \to 0$ their contributions cancel, so and the outer curve goes to zero as discussed hence we prove the claim.

Definition. (Local degree) Let $a \in \mathbb{C}$, and $f : B(a, r) \to \mathbb{C}$ holomorphic and constant. The *local degree* of $\deg_{z=a}(f)$ of f at a is the order of the zero at a of $f(z) - f(a)$.

Lemma. If $\gamma(t) = a + \varepsilon e^{it}$ where $0 \leq t \leq 2\pi$ and $\varepsilon > 0$ sufficiently small, then

$$\deg_{z=a} f = I(f \circ \gamma, f(a)).$$

Proof. By the identity theorem since g is non-constant there exists a $\delta > 0$ such that g does not vanish on $B(a, \delta)^*$. Take $0 < \varepsilon < \delta$. Then $I(f \circ \gamma, f(a))$ is well-defined since $f(a) \notin \mathrm{im}(f \circ g)$. Now this is the argument principle using $I(g \circ \gamma, 0) = I(f \circ \gamma, f(a))$. □

Theorem. (Local mapping degree) Let $f : B(a, r) \to \mathbb{C}$ be holomorphic, non-constant and with $\deg_{z=a}(f) = k > 0$. Then for all $\rho > 0$ sufficiently small, there exists a $\varepsilon > 0$ such that if

$$0 < |w - f(a)| < \varepsilon$$

there $f(z) = w$ has exactly k simple roots in $B(a, \rho)$ (i.e. roots with degree 1).

Proof. Pick $\rho > 0$ such that $g(z) = f(z) - f(a)$ and $g'(z) = f'(z)$ are both non-vanishing in $B(a, \rho)^*$ by the principle of isolated zeros. Now let $\gamma(t) = a + \rho e^{2\pi t}$ for $0 \leq 1 \leq 1$. Then $f(a) \notin \mathrm{im}(f \circ \gamma)$. So there exists a $\varepsilon > 0$ such that $\mathrm{im}(f \circ \gamma) \cap B(f(a), \varepsilon) = \emptyset$. Since $f \circ \gamma$ has a closed bounded image. If we take $w \in B(f(a), \varepsilon)$, we count solutions to $f(z) = w$ inside $B(a, \rho)$ using the argument principle.

The number of zeros of $f(z) - w$ in $B(a, \rho)$ is equal to $(I(f \circ \gamma, w) = I(f \circ \gamma, f(a))$. Using that $I(f \circ \gamma, \circ)$ depends only on components of $\mathbb{C} \backslash \mathrm{im}(f \circ \gamma)$ containing $\cdot$. Now this is just $\deg_{z=a}(f) = k$.

So we have k zeros with multiplicity, now it's enough to show each root is simple. Since f' is non-vanishing in $B(a, \rho)$, each root has be simple, so we have k distinct simple roots. $\square$

Corollary. (Open mapping theorem) Let U be a domain and $f : U \to \mathbb{C}$ be holomorphc and non-constant. Then f is an open map.

Proof. Suppose $V \subseteq U$ open. Let $z \in V$ hence there exists a $\rho > 0$ such taht $B(z, \rho) \subseteq V$. Hence there exists a ε such that $B(\varepsilon, f(z)) \subseteq f(B(z, \rho)) \subseteq f(U) \subseteq \mathbb{C}$ by the local mapping degree theorem, given that $k \geq 1$ since f is non-constant. $\square$

Corollary. Let $U \subseteq \mathbb{C}$ be proper and simply connected. Then there exists a holomorphic injection from U to $\mathbb{D}$.

Remark. If $U = \mathbb{C}$ then the conclusion would be definitely false by Liouville's theorem.

Proof. Let $q \in \mathbb{C} \setminus U$. Let $\phi(z) = z - q$. So on U, ϕ is non-vanishing. Since U is simply connected, there exists a branch of the logarithm on U, so we can solve the equation $\phi = h^2$ for h holomorphic on U since the branch of the logarithm says that there exists a g such that $\phi = e^g$, and then we can take $h = e^{g/2}$.

Pick some point $y \in h(U)$. The open mapping theorem, says that there exists a ε ball centred at y. Now since ϕ is injective, this implies that $h(U) \cap B(-y, \varepsilon) = \emptyset$ since if $h(z_1) = \pm h(z_2)$ then $z_1 = z_2$. Let's consider the map

$$z \to \frac{\varepsilon}{2(h(z) + y)}$$

which is holomorphic on U and the above shows that this has image in $\mathbb{D}$. $\square$

Definition. (Local uniform convergence) Let $U \subset \mathbb{C}$ be open and let $f_n : U \to \mathbb{C}$ be holomorphic. We say that $\{f_n\}$ converges *locally uniformly* to f on U if $\forall a \in U$, there is a $B(a, r) \subseteq U$ such that $f_n \to f$ uniformly on $B(a, r)$.

Example. Let $f_n(z) = z^n$ on $B(0, 1)$. Then $f_n \to f$ converges pointwise and indeed locally uniformly to zero, but not uniformly on $B(0, 1)$. If $a \in B(0, 1)$ and we pick a r such that $|a| + r < 1$. Then there exists a $\rho < 1$ such that $B(a, r) \subset \overline{B(0, \rho)} \subset B(0, 1)$.

Now notice that $|f_n(z)| < \varepsilon \iff |z| < \varepsilon^{1/n}$. If we pick a N sufficiently large such that $\rho < \varepsilon^{1/N} < 1$ then we see that $\{f_n\}$ converges to zero uniformly on $\overline{B(0, \rho)}$.

Now using this we can make a corollary to Morera's theorem.

Corollary. If U is a domain and $f_n \to f$ locally uniformly on U with f_n holomorphic for all n, then f is holomorphic and $f_n' \to f'$ locally uniformly.

Remark. We've proved that $f_n' \to f'$ *pointwise*, but now we can prove this slightly stronger statement.

Proof. For $a \in U$ pick $r > 0$ such that $f_n \to f$ uniformly on $\overline{B(a, r)} \subset U$. For a closed contour γ in $B(a, r)$ we have that

$$\int_\gamma f_n = 0$$

hence we have that

$$\int_\gamma f = 0$$

for all γ, so by Morera's theorem f is holomorphic. By Taylor's theorem we have that

$$|f_n'(w) - f'(w)| = \frac{1}{2\pi i}\left|\int_{|z-a|=\rho} \frac{f_n(z) - f(z)}{(z-w)^2}\,\mathrm{d}z\right|$$

for $0 < \rho < r$ and $w \in B(a, \rho)$. If $|w-a| \leq \frac{\rho}{2}$, $|f_n'(w) - f'(w)| \leq \frac{1}{2\pi i} \cdot 2\pi\rho \sup_{|z-a|=\rho} |f(z) - f_n(z)| \to 0$ by uniform convergence of $f_n \to f$ and bound by some ε uniformly in w since the RHS is independent of w hence the convergence is locally uniform. □

Proposition. Suppose U is a domain $f_n : U \to \mathbb{C}$ be holomorphic and injective with $f_n \to f$ uniformly. Then f is injective or constant.

Proof. Suppose for contradiction that f is non-constant and non-injective, so there exists $z_1, z_2 \in U$ such that $f(z_1) = f(z_2) = a$. U is a domain so path-connected, hence we can joint z_1, z_2 by a polygonal path in U created by line segments parallel to the axes. Construct a closed curve γ in U bounding a domain in U containing z_1 and z_2. *Wlog* we can assume that $f(z) \neq 0$ for all z on γ since by the identity theorem $f^{-1}(a)$ consists of isolated points (since they are zeros of the holomorphic function $f(z) - a$). Now using the argument principle, we have that

$$1 \geq \frac{1}{2\pi i}\int_\gamma \frac{f_n'(z)}{f_n(z) - a}\,\mathrm{d}z$$

since f_n is injective for all n. This converges to

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z) - a}\,\mathrm{d}z$$

since $f_n \to f$ and $f_n' \to f'$ locally uniformly and hence on the closed bounded set bounded by γ. But this integral is at least 2 since z_1, z_2 are both zeros of $f(z) - a$ which is a contradiction. □

We will now quote the following theorem without proof.

Theorem. (Montel's theorem) Let $\mathcal{F}$ be a family of holomorphic functions on a domain U. Assume that $\mathcal{F}$ is uniformly bounded, so there exists some $R > 0$ such that $\mathrm{im}(f) \subset B(0, R)$ for all $f \in \mathcal{F}$. Then if (f_j) is any sequence in $\mathcal{F}$ is has a locally uniformly convergent subsequence.

Remark. A fundamental theorem in real analysis (Arzelà-Asocli Theorem) says that if $\{f_n : [a, b] \to \mathbb{R}\}$ are uniformly bounded and uniformly equicontinuous so for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|f_n(x) - f_n(y)| < \varepsilon$$

whenever $|x - y| < \delta$ for all functions f_n in the sequence. So δ may depend on ε but not on x, y, n. Then there exists a subsequence converging uniformly.

Theorem. (Riemann Mapping Theorem) Suppose $U \neq \mathbb{C}$ is a simply connected domain. Then there exists a conformal equivalence from U to $B(0,1)$.

Proof. Let

$$\mathcal{F} = \{f : U \to B(0,1) : f \text{ holomorphic, inejctive, and take } z_0 \to 0\}$$

where we pick $z_0 \in U$. We will prove the theorem in three steps

(i) $\mathcal{F}$ is non-empty;

(ii) There exists some $f_\infty \in \mathcal{F}$ such that $|f'_\infty(z_0)|$ is the maximal value this achieves over all $f \in \mathcal{F}$;

(iii) f_∞ is surjective and defines the required conformal isomorphism.

We know (i) to be true from the open mapping theorem. We showed there existed some $f : U \to B(0,1)$ holomorphic and injective. If this f wasn't such that $f(z_0) \neq 0$ we can compose with a Möbius map to send $f(z)$ to 0.

For (ii), pick f_n in $\mathcal{F}$ such that $|f'_n(z_0)| \to \sup_{f\in\mathcal{F}} |f'(z_0)|$. $\mathcal{F}$ is a uniformly bounded family. so there exists a subsequence $f_{n_j} \to f_\infty$ locally uniformly by Montel. Then $|f'_\infty(z_0)| = \max_{f\in\mathcal{F}}(f'(z_0)) > 0$. So f_∞ is non-constant, hence it's injective and holomorphic so it's in $\mathcal{F}$.

It remains to show that f_∞ is surjective for (iii). Suppose that f_∞ is not into $B(0,1)$, say $\alpha \in B(0,1) \setminus f_\infty(U)$ then consider

$$\sigma_\alpha(z) = \frac{z-\alpha}{z\overline{\alpha}-1} \in \mathcal{M}(B(0,1))$$

sand $\sigma_\alpha \circ f_\infty(U)$ is a simply connected domain $B(0,1)$ not containing 0. So there exists a holomorphic branch of $\mathrm{Log}(\sigma_\alpha \circ f_\infty)$ on U, hence this function has a square root $s(z)$. So let $h(z) = \sigma_{s(\alpha)} \circ s \circ (\sigma_\alpha \circ f_\infty) : U \to B(0,1)$. Since $s^2 = \sigma_\alpha \circ f_\infty$ then one checks that $h \in \mathcal{F}$. Write $f_\infty = \Phi \circ h$ where $\Phi = \sigma_\alpha^{-1} \circ (\text{squaring}) \circ \sigma_{s(\alpha)}^{-1}$. Now $\Phi : B(0,1) \to B(0,1)$ is holomorphic with $\Phi(0) = 0$ and not a rotation. So we can apply Schwarz's Lemma from Example Sheet 2. Hence $|\phi'(0)| < 1$, so $|f'_\infty(z_0)| < |h'(z_0)|$ which is a contradiction to the construction of f_∞. Hence we have a contradiction, so f_∞ is surjective onto $B(0,1)$. It's conformal everywhere and by the inverse function theorem it is an isomorphism so $U \cong B(0,1)$. $\square$

Remark. This proof feels very opaque. The complexity is hidden amongst many theorem and it's not clear what the map looks like. Almost always these maps are not expressible in elementary functions.